Tutorial on Visual Odometry

Davide Scaramuzza

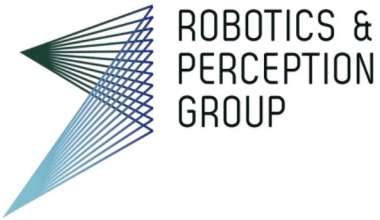
University of Zurich

Robotics and Perception Group

http://rpg.ifi.uzh.ch/

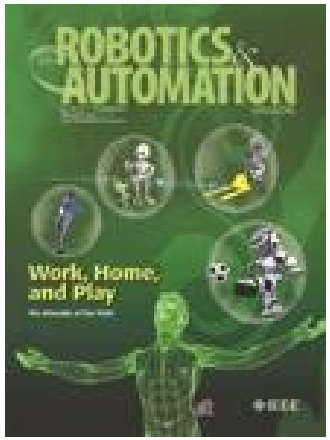


University of Zurich[UZH]



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

References





- Scaramuzza, D., Fraundorfer, F., **Visual Odometry: Part I** - The First 30 Years and Fundamentals, IEEE Robotics and Automation Magazine, Volume 18, issue 4, 2011.

- Fraundorfer, F., Scaramuzza, D., **Visual Odometry: Part II** - Matching, Robustness, and Applications, IEEE Robotics and Automation Magazine, Volume 19, issue 1, 2012.

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

What is Visual Odometry (VO) ?

VO is the process of incrementally estimating the pose of the vehicle by examining the changes that motion induces on the images of its onboard cameras

input

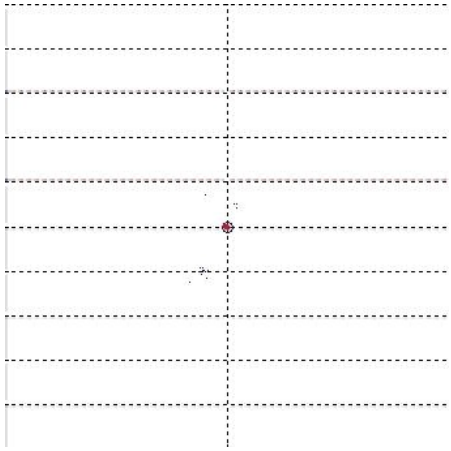


Image sequence (or video stream)
from one or more cameras attached to a moving vehicle





output



$$R_0, R_1, \ldots, R_i$$

$$t_0, t_1, \ldots, t_i$$

Camera trajectory (3D structure is a plus):

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Assumptions

- Sufficient illumination in the environment
- Dominance of static scene over moving objects
- Enough texture to allow apparent motion to be extracted
- Sufficient scene overlap between consecutive frames











Is any of these scenes good for VO? Why?

Copyright © Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Why VO ?

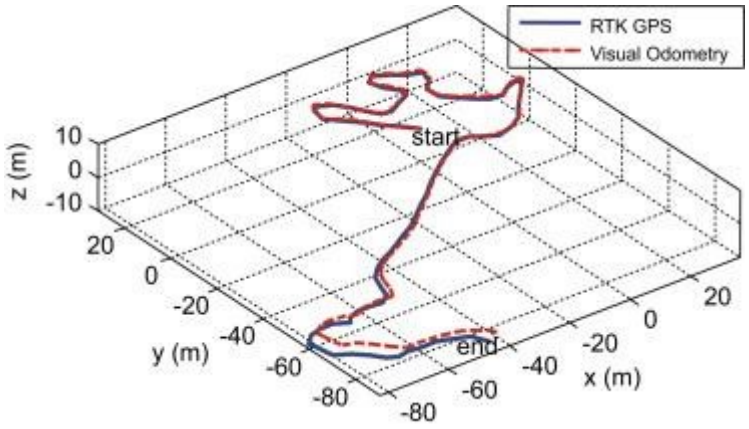
- Contrary to wheel odometry, VO is not affected by wheel slip in uneven terrain or other adverse conditions.



- More accurate trajectory estimates compared to wheel odometry
  (relative position error 0.1% – 2%)

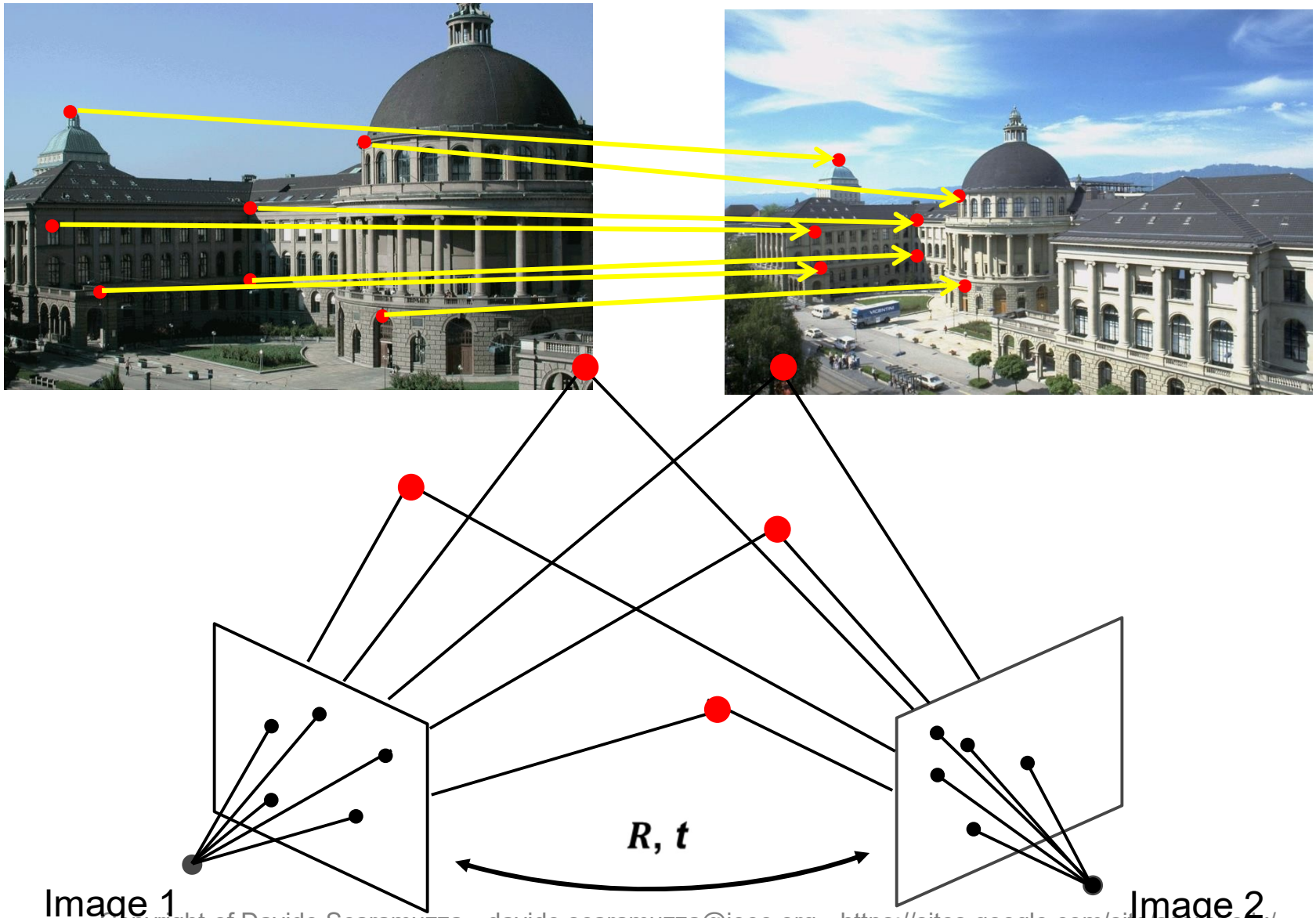
- VO can be used as a complement to
  - wheel odometry
  - GPS
  - inertial measurement units (IMUs)
  - laser odometry

- In GPS-denied environments, such as underwater and aerial, VO has utmost importance



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

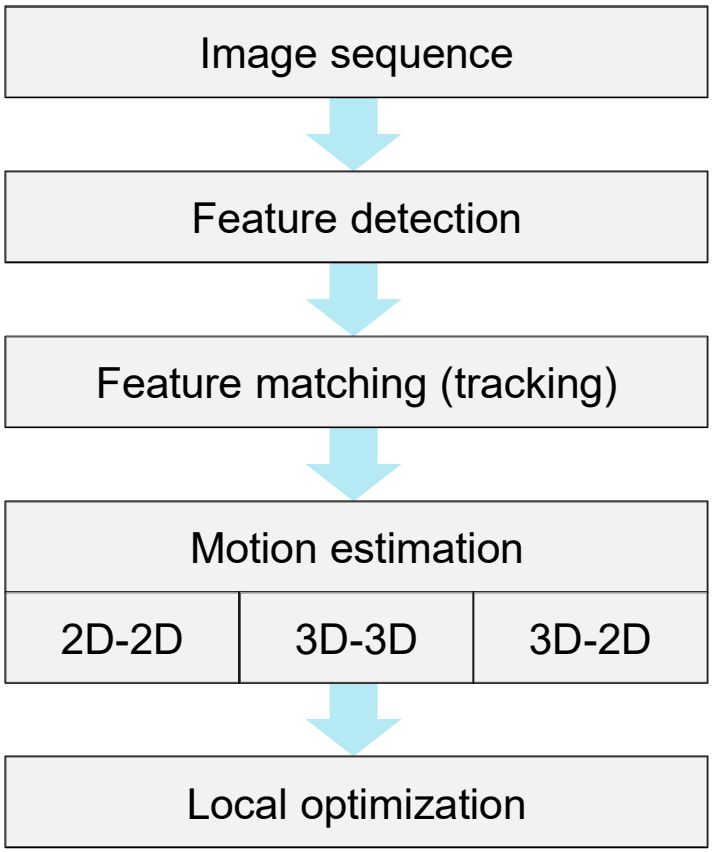
Working Principle

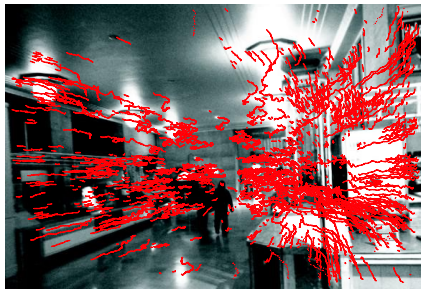


Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

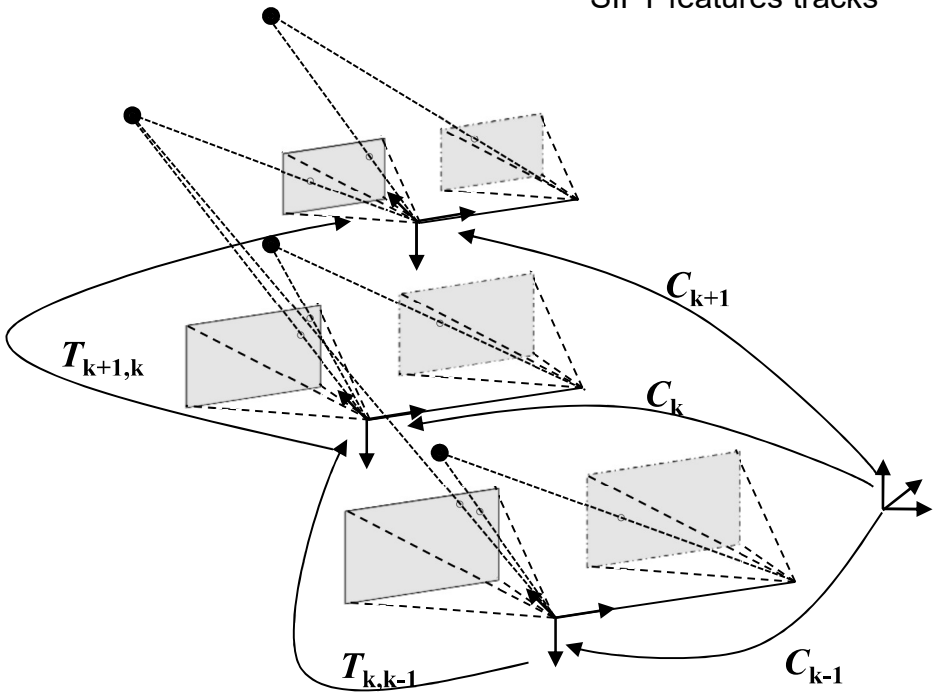
VO Flow Chart

- VO computes the camera path incrementally (pose after pose)





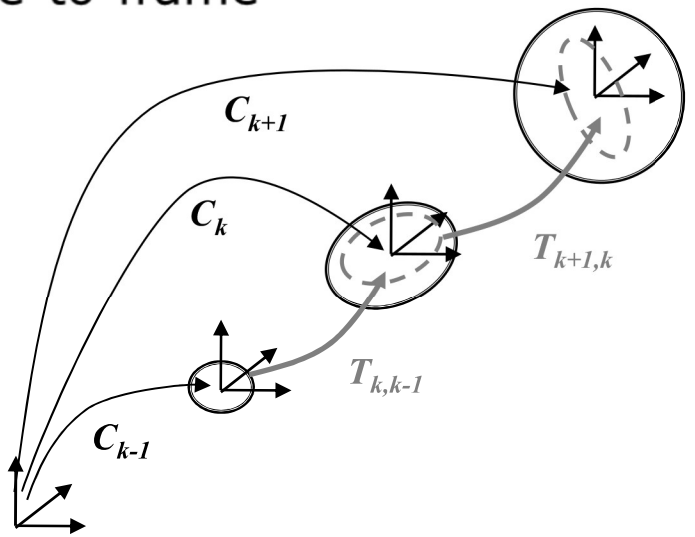
SIFT features tracks



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

VO Drift

- The errors introduced by each new frame-to-frame motion accumulate over time
- This generates a drift of the estimated trajectory from the real path



The uncertainty of the camera pose at C_k is a combination of the uncertainty at C_{k-1} (black solid ellipse) and the uncertainty of the transformation $T_{k,k-1}$ (gray dashed ellipse)

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

VO or Structure from Motion (SFM) ? (1/2)

- SFM is more general than VO and tackles the problem of 3D reconstruction of both the structure and camera poses from unordered image sets

- The final structure and camera poses are typically refined with an offline optimization (i.e., bundle adjustment), whose computation time grows with the number of images

This video can be seen at
http://youtu.be/kxtQqYLRaSQ

Reconstruction from 3 million images from Flickr.com
Cluster of 250 computers, 24 hours of computation!
Paper: “Building Rome in a Day”, ICCV’09

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

VO or Structure from Motion (SFM) ? (2/2)

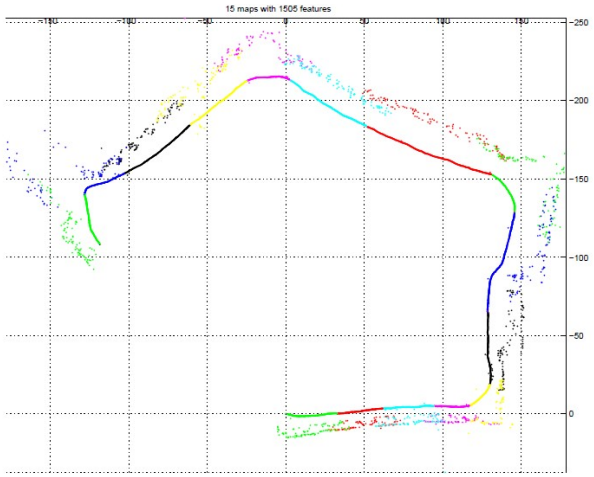
- VO is a particular case of SFM
- VO focuses on estimating the 3D motion of the camera sequentially (as a new frame arrives) and in real time.
- Bundle adjustment can be used (but it's optional) to refine the local estimate of the trajectory
- Terminology: sometimes SFM is used as a synonym of VO

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

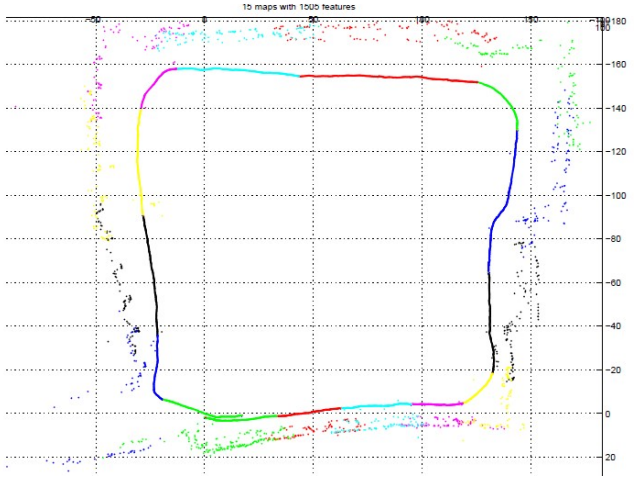
VO vs. Visual SLAM (1/2)

- The goal of SLAM in general is to obtain a global, consistent estimate of the robot path. This is done through identifying loop closures. When a loop closure is detected, this information is used to reduce the drift in both the map and camera path (global bundle adjustment).

- Conversely, VO aims at recovering the path incrementally, pose after pose, and potentially optimizing only over the last m poses path (windowed bundle adjustment)



Before loop closing



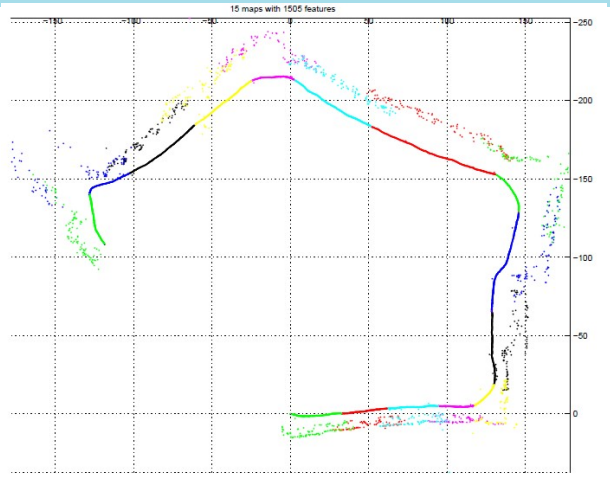
After loop closing

Image courtesy of Clemente et al. RSS'07

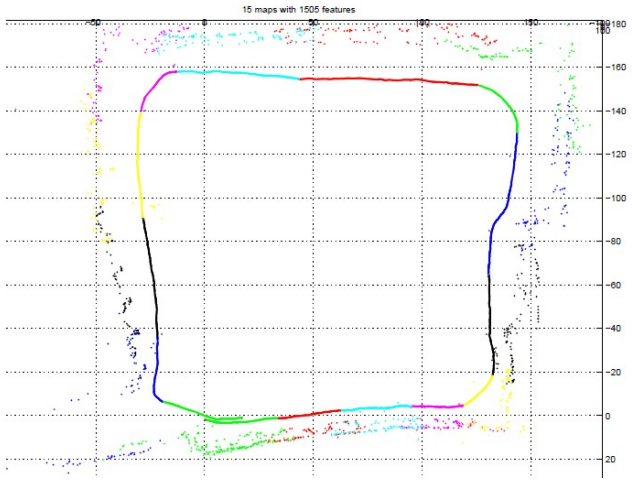
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

VO vs. Visual SLAM (2/2)

- VO only aims to the local consistency of the trajectory
- SLAM aims to the global consistency of the trajectory and of the map
- VO can be used as a building block of SLAM
- VO is SLAM before closing the loop!
- The choice between VO and V-SLAM depends on the tradeoff between performance and consistency, and simplicity in implementation.
- VO trades off consistency for real-time performance, without the need to keep track of all the previous history of the camera.



Visual odometry



Visual SLAM

Image courtesy of Clemente et al. RSS'07

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

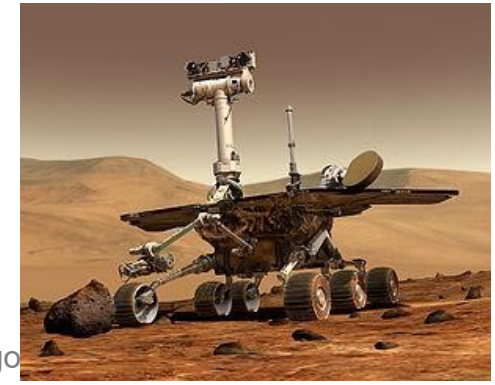
Outline

- Brief history of VO
- Problem formulation
- Camera modeling and calibration
- Motion estimation
- Robust estimation
- Error propagation
- Camera-pose optimization (bundle adjustment)
- Discussion

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Brief history of VO

- **1996**: The term VO was coined by Srinivasan to define motion orientation in honey bees.
- **1980**: First known stereo VO real-time implementation on a robot by Moraveck PhD thesis (NASA/JPL) for Mars rovers using a sliding camera. Moravec invented a predecessor of Harris detector, known as Moravec detector
- **1980 to 2000**: The VO research was dominated by NASA/JPL in preparation of 2004 Mars mission (see papers from Matthies, Olson, etc. From JPL)
- **2004**: VO used on a robot on another planet: Mars rovers Spirit and Opportunity
- **2004**. VO was revived in the academic environment by Nister «Visual Odometry» paper.
  The term VO became popular.



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.go

Outline

- Brief history of VO
- Problem formulation
- Camera modeling and calibration
- Motion estimation
- Robust estimation
- Error propagation
- Camera-pose optimization (bundle adjustment)
- Discussion

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Problem Formulation

- An agent is moving through an environment and taking images with a rigidly-attached camera system at discrete times k

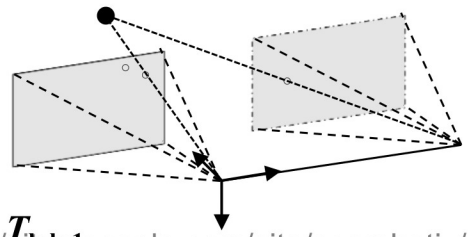
- In case of a monocular system, the set of images taken at times k is denoted by

$$I_{0:n} = \{I_0, \ldots, I_n\}$$

- In case of a stereo system, there are a left and a right image at every time instant, denoted by

$$I_{l,0:n} = \{I_{l,0}, \ldots, I_{l,n}\},$$
$$I_{r,0:n} = \{I_{r,0}, \ldots, I_{r,n}\}$$

- In case of a stereo system, without loss of generality, the coordinate system of the left camera can be used as the origin



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

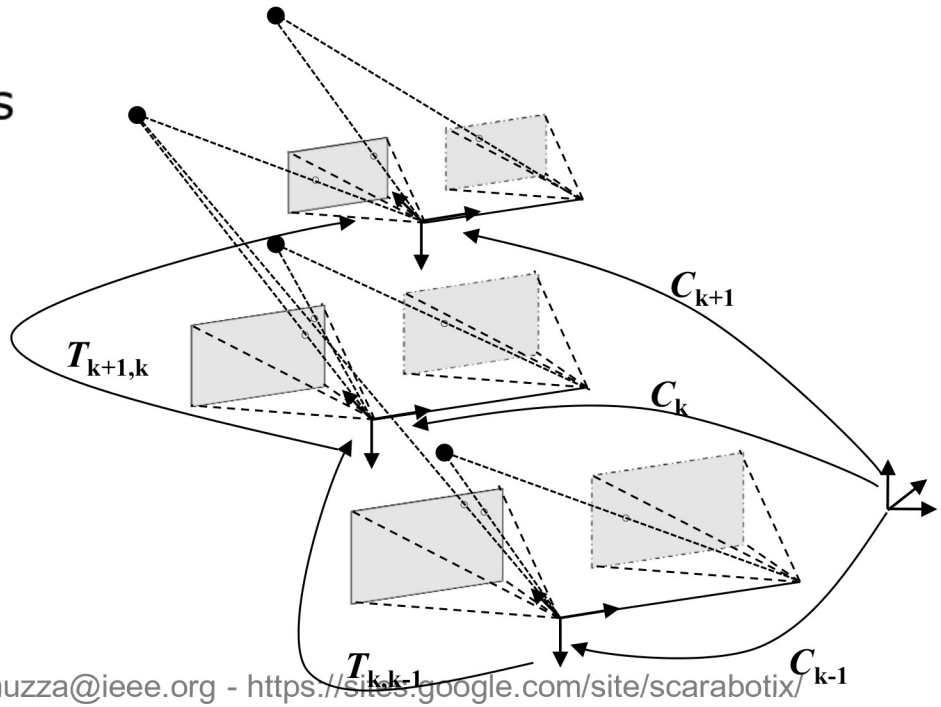
Problem Formulation

- Two camera positions at adjacent time instants $k-1$ and k are related by the rigid body transformation

$$T_k = \begin{bmatrix} R_{k,k-1} & t_{k,k-1} \\ 0 & 1 \end{bmatrix}$$

- The set $T_{0:n} = \{T_1, \ldots, T_n\}$ contains all subsequent motions all subsequent motions.

- Finally, the set of camera poses $C_{0:n} = \{C_0, \ldots, C_n\}$ contains the transformations of the camera with respect to the initial coordinate frame at $k = 0$



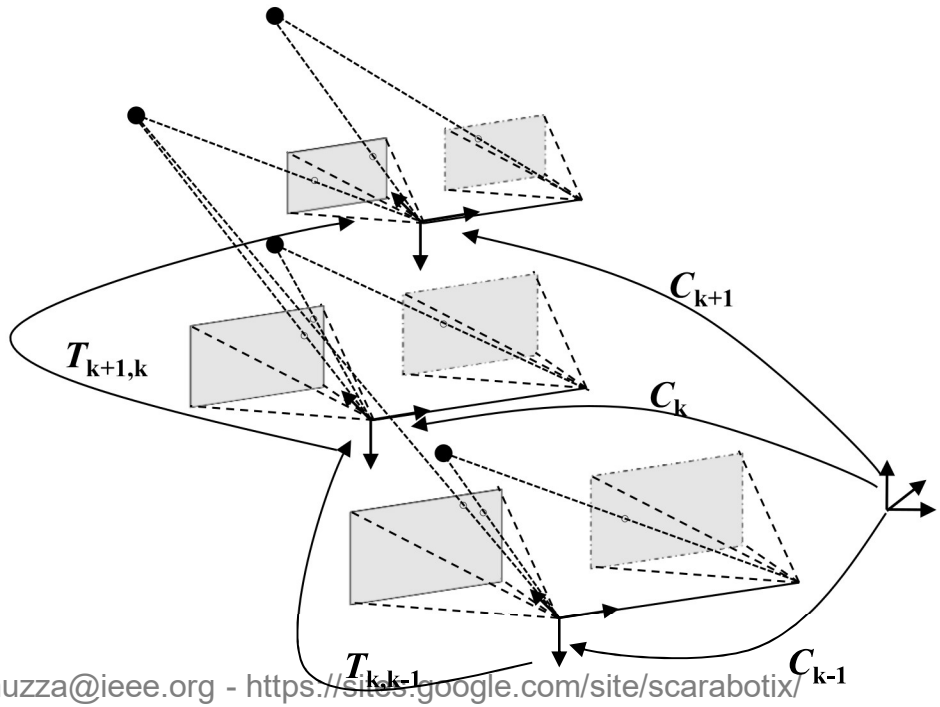
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Problem Formulation

- The current pose C_n can be computed by concatenating all the transformations T_k, $k = 1 \dots n$, and, therefore,

$$C_n = C_{n-1} T_n$$

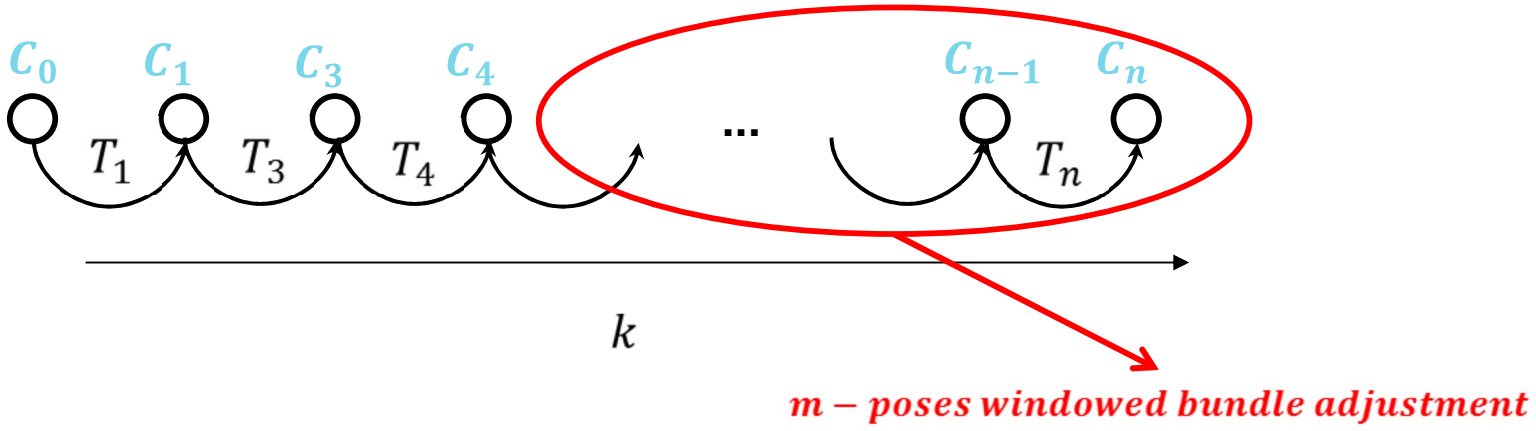
with C_0 being the camera pose at the instant $k = 0$, which can be set arbitrarily by the user



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Problem Formulation

- The main task in VO is to compute the relative transformations T_k from the images I_k and I_{k-1} and then to concatenate the transformations to recover the full trajectory $C_{0:n}$ of the camera.
- This means that VO recovers the path incrementally, pose after pose.
- An iterative refinement over last *m* poses can be performed after this step to obtain a more accurate estimate of the local trajectory.



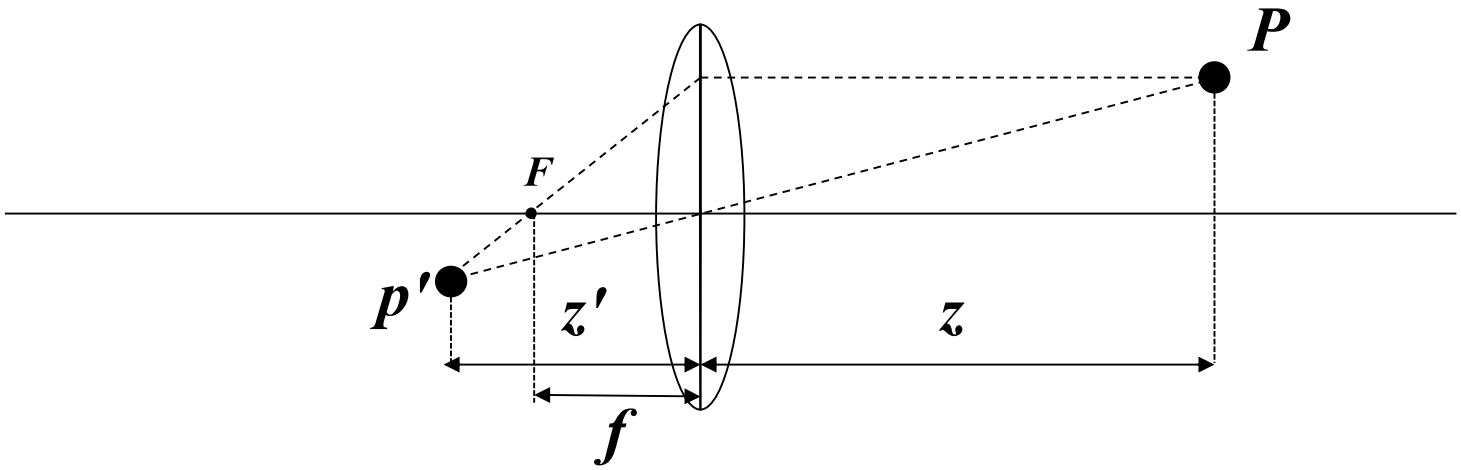
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Outline

- Brief history of VO
- Problem formulation
- Camera modeling and calibration
- Motion estimation
- Robust estimation
- Error propagation
- Camera-pose optimization (bundle adjustment)
- Discussion

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

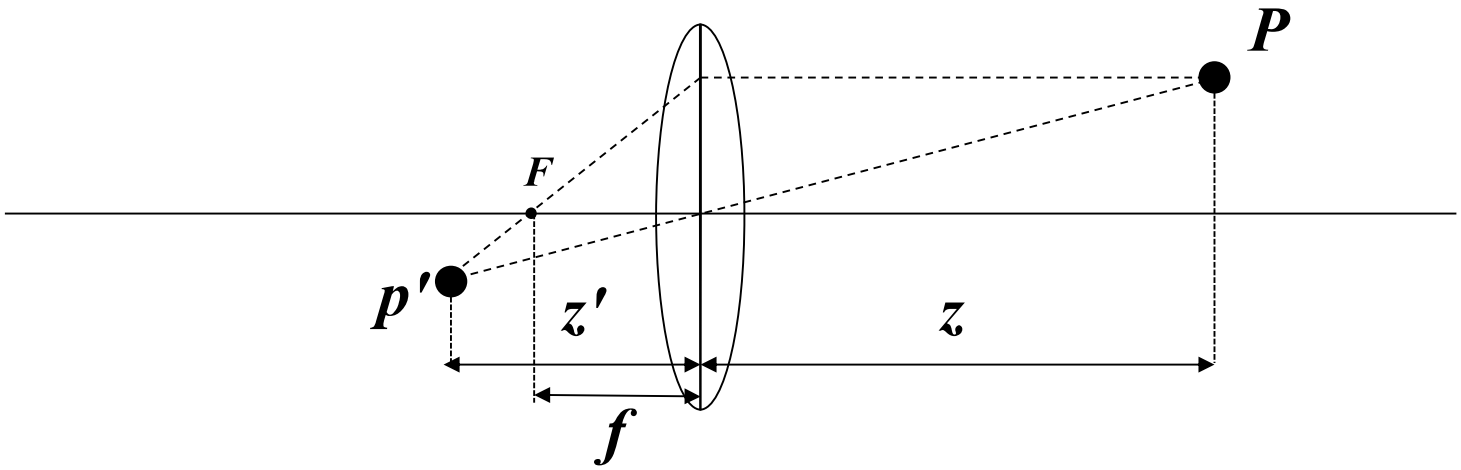
Thin Lens Model



$$\frac{1}{z}+\frac{1}{z'}=\frac{1}{f}$$

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

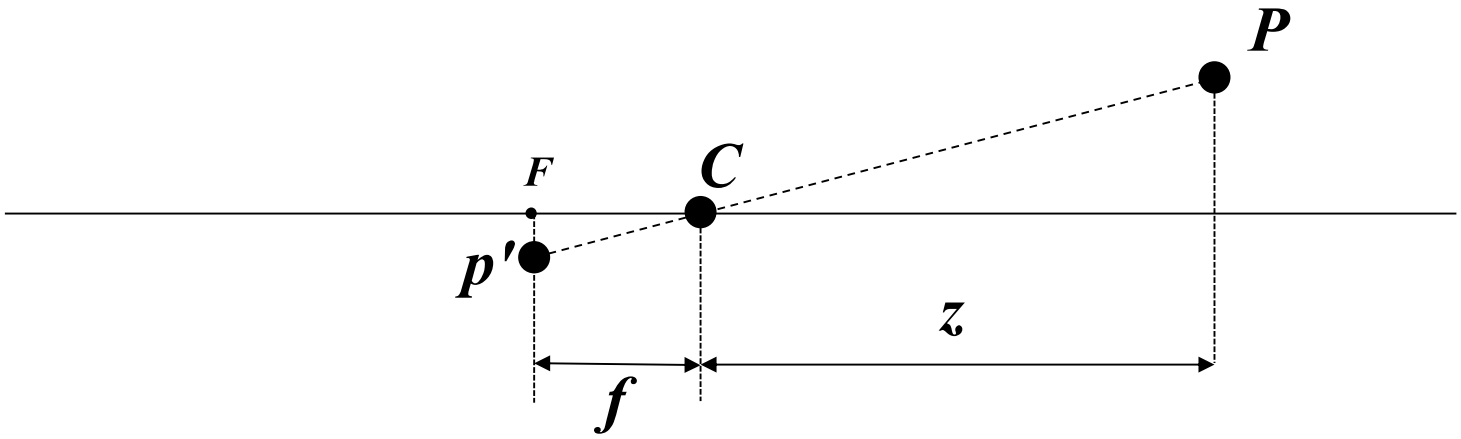
Thin Lens Model



$$z >> z' \quad \Rightarrow \frac{1}{z'} \approx \frac{1}{f} \quad \Rightarrow z' = f$$

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

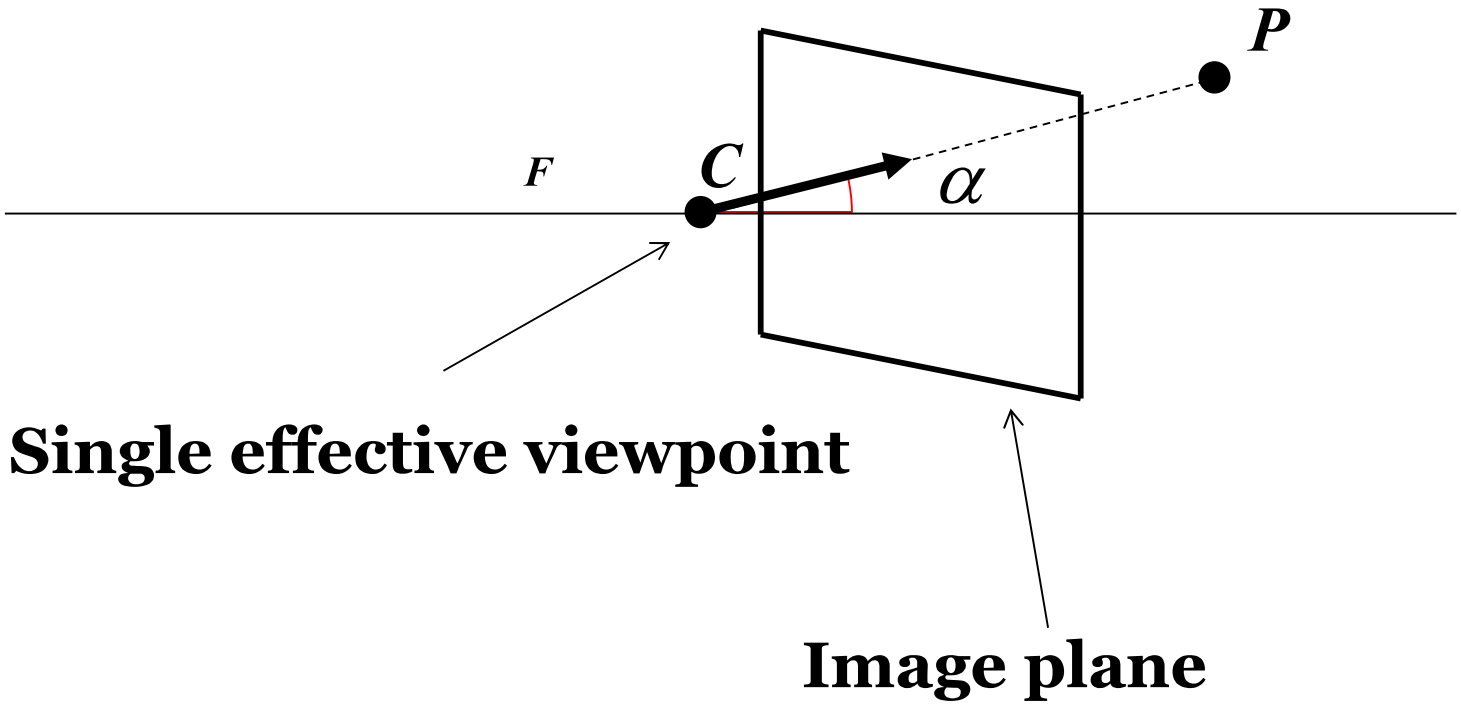
The Pin-Hole Approximation



$$z >> z' \quad \Rightarrow \frac{1}{z'} \approx \frac{1}{f} \quad \Rightarrow z' = f$$

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

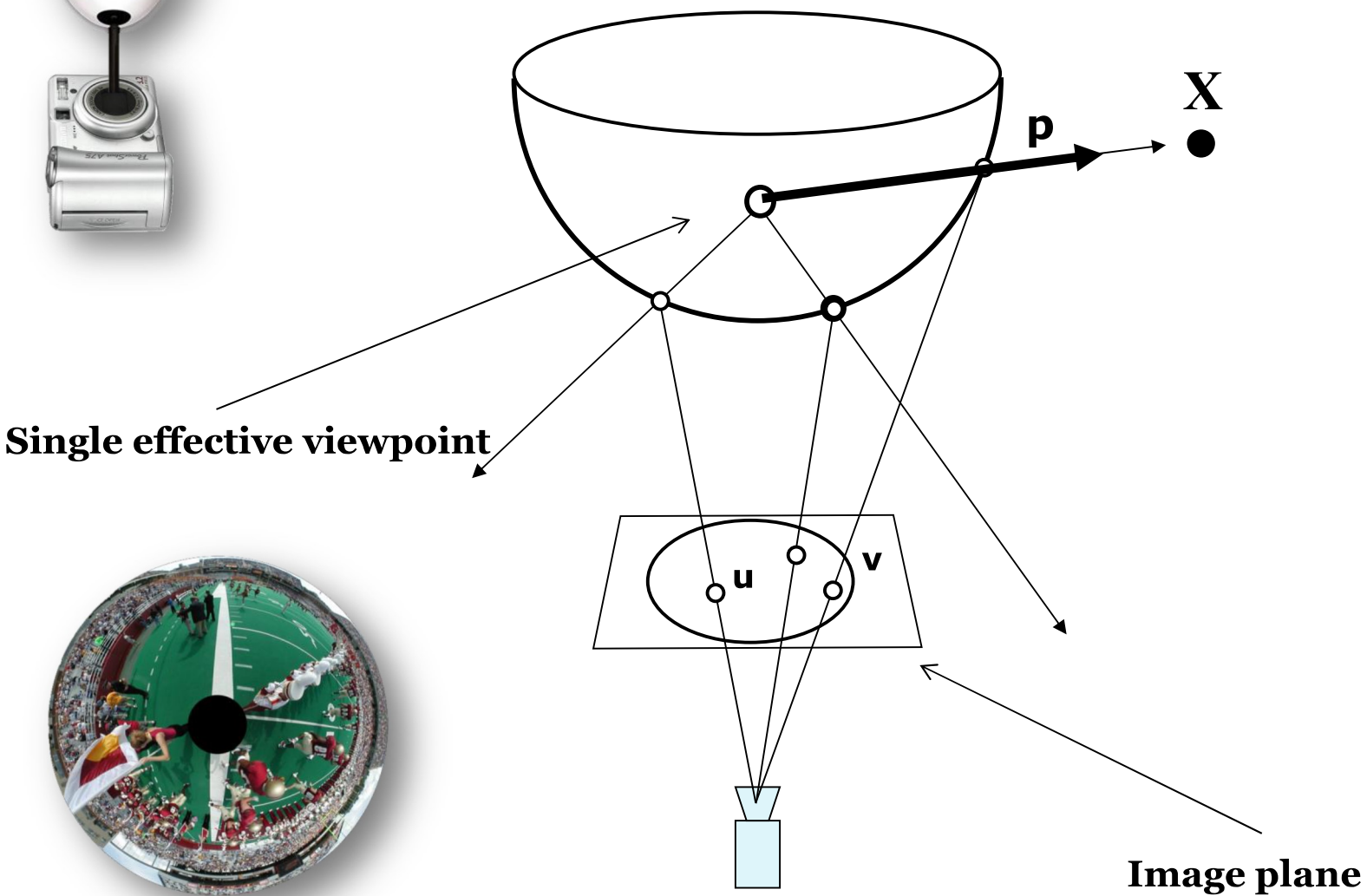
The Pin-Hole Approximation



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

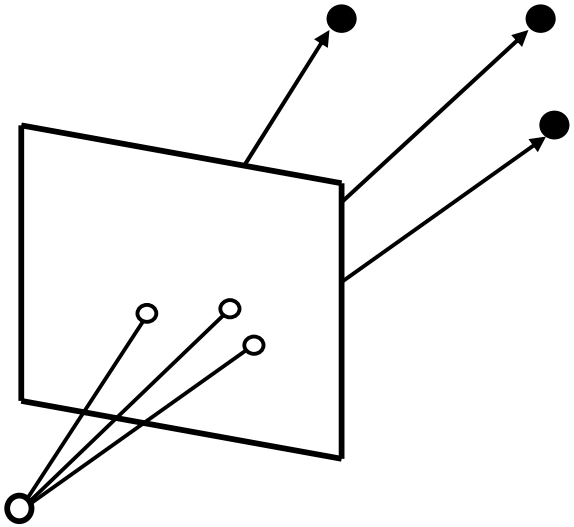
Omnidirectional Camera Model

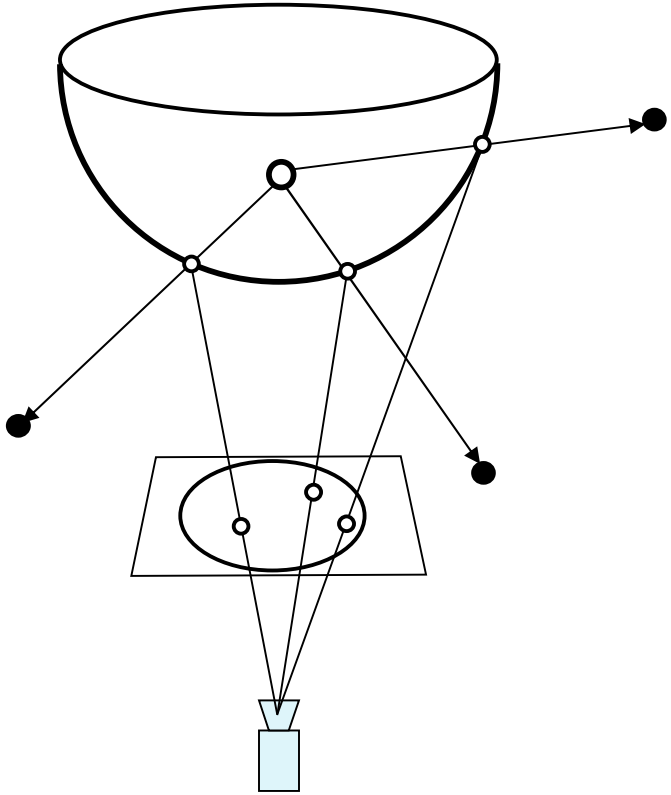




Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

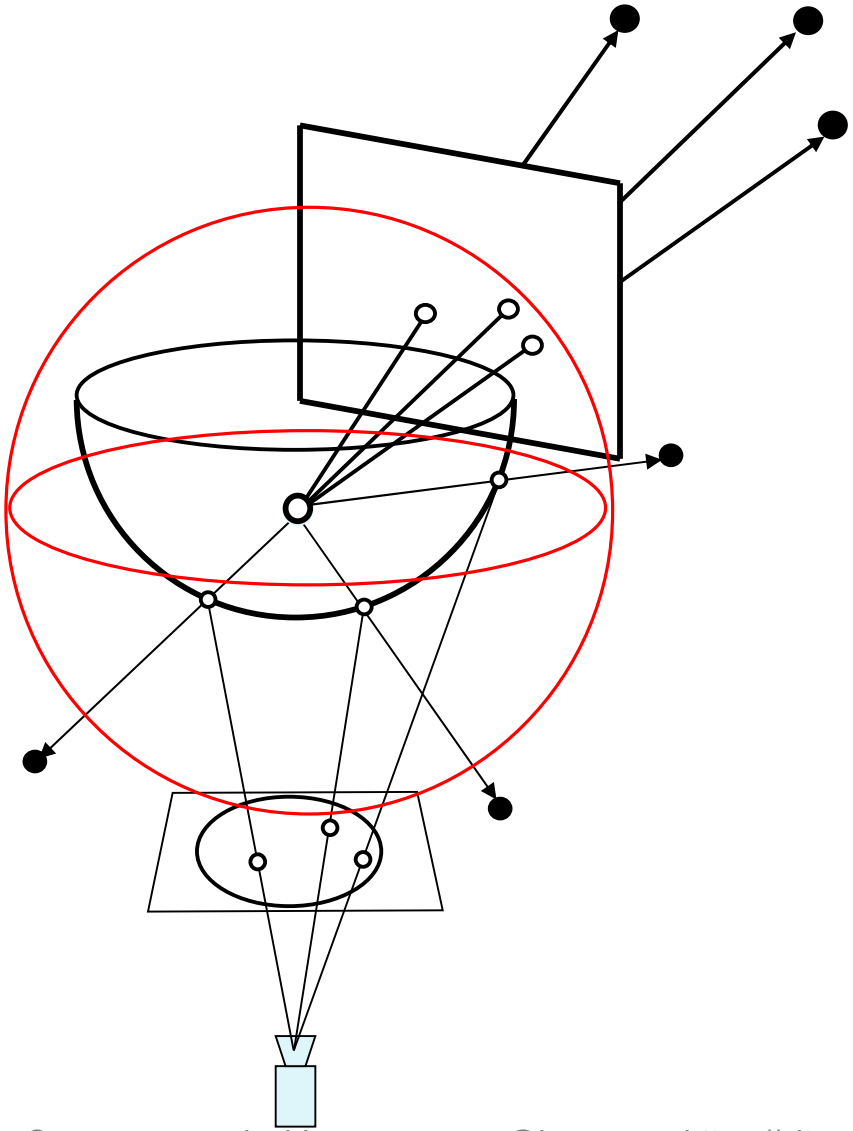
Equivalence between Perspective and Omnidirectional Model





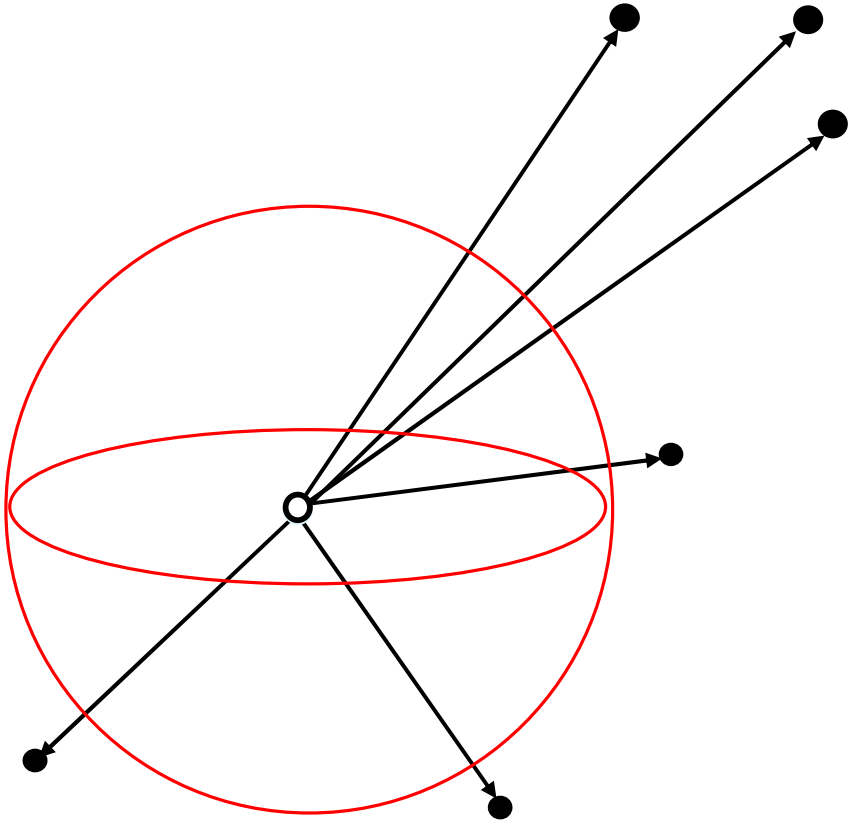
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Equivalence between Perspective and Omnidirectional Model



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Equivalence between Perspective and Omnidirectional Model: The Spherical Model



- Always possible after the camera has been calibrated!

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Equivalence between Perspective and Omnidirectional Model: The Spherical Model

- For convenience, points are projected on the unit sphere. Why?
- In the perspective case, is it better to use the perspective or the spherical model?

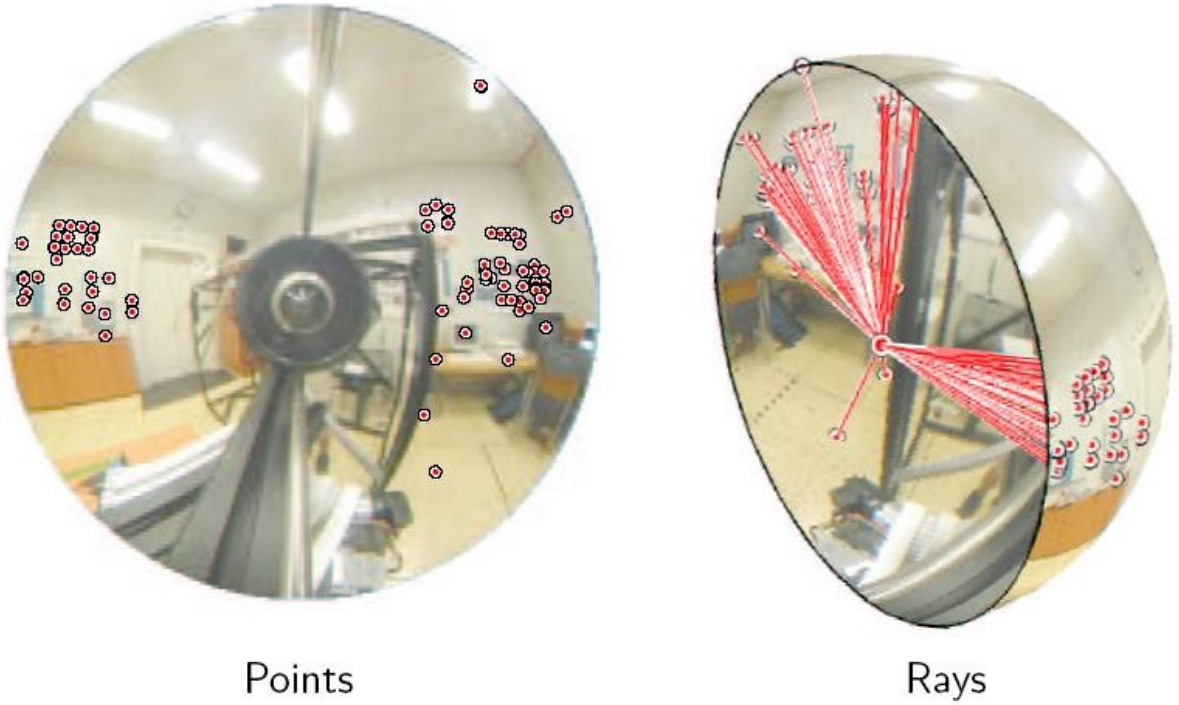
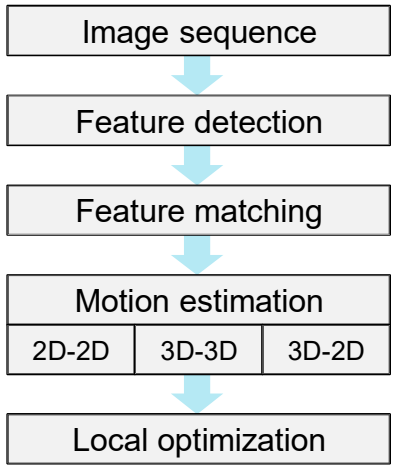


Image courtesy of Micusik & Pajdla, ACCV'04

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Outline

- Brief history of VO
- Problem formulation
- Camera modeling and calibration
- Motion estimation
- Robust estimation
- Error propagation
- Camera-pose optimization (bundle adjustment)
- Discussion



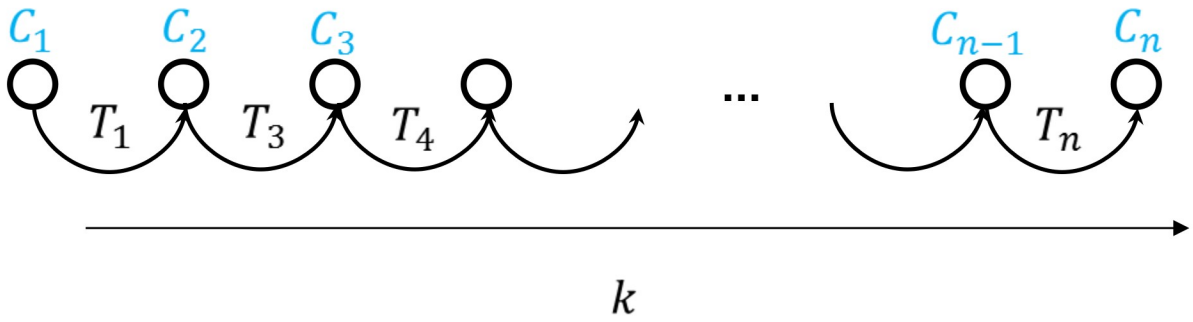
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Motion Estimation

- Motion estimation is the core computation step performed for every image in a VO system
- It computes the camera motion T_k between the previous and the current image:

$$T_k = \begin{bmatrix} R_{k,k-1} & t_{k,k-1} \\ 0 & 1 \end{bmatrix}$$

- By concatenation of all these single movements, the full trajectory of the camera can be recovered



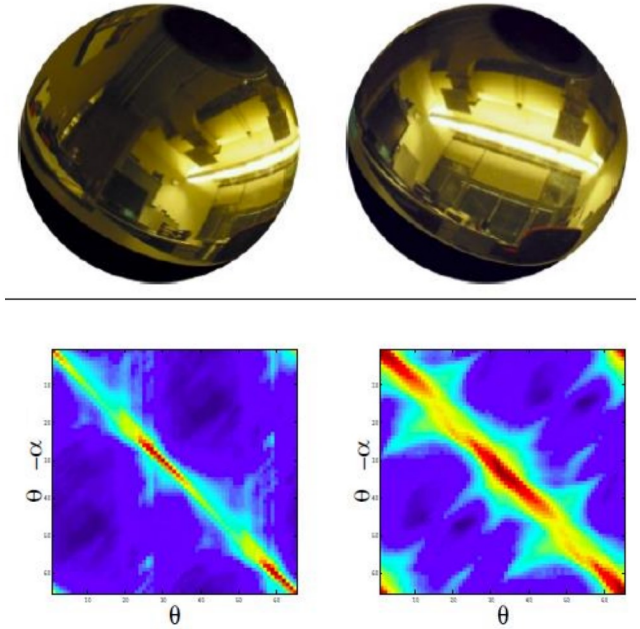
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

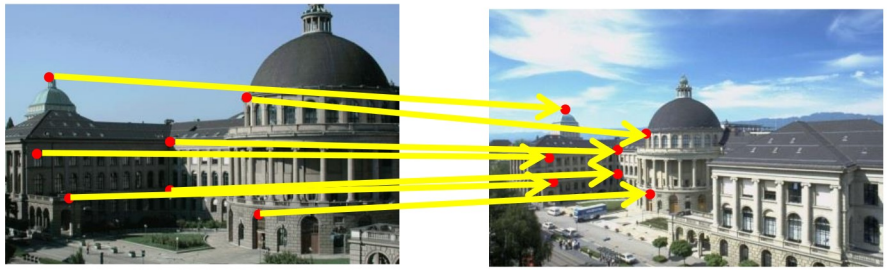
Appearance-based or Featured-based ?

- There are two main approaches to compute the relative motion T_k

- Appearance-based methods use the intensity information of all the pixels in the two input images

- Feature-based methods only use salient and repeatable features extracted (or tracked) across the images





Makadia et al. «Correspondence-free structure from motion», IJCV'07

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Appearance-based or Featured-based ?

- Global methods are less accurate than feature-based methods and are computationally more expensive.

- Feature-based methods require the ability to match (or track) robustly features across frames but are faster and more accurate than global methods. Therefore, most VO implementations are feature based.

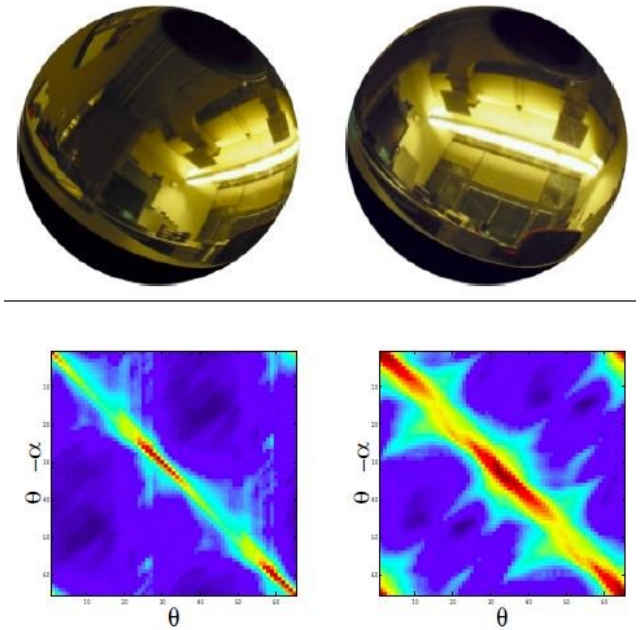
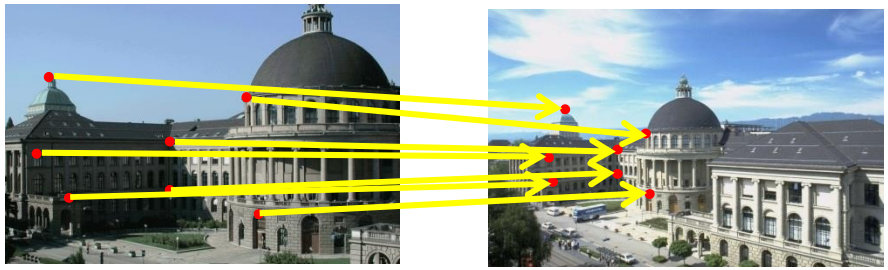


Image courtesy of Makadia et al., IJCV'07



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Motion Estimation

Motion estimation		
2D-2D	3D-3D	3D-2D

- Depending on whether the feature correspondences f_{k-1} and f_k are specified in 2D or 3D, there are three different cases:

- **2D-to-2D**: both f_{k-1} and f_k are specified in 2D image coordinates

- **3D-to-3D**: both f_{k-1} and f_k are specified in 3D To do this, it is necessary to triangulate 3D points at each time instant, for instance, by using a stereo camera system

- **3D-to-2D**: f_{k-1} are specified in 3D and f_k are their corresponding 2D reprojections on the image I_k

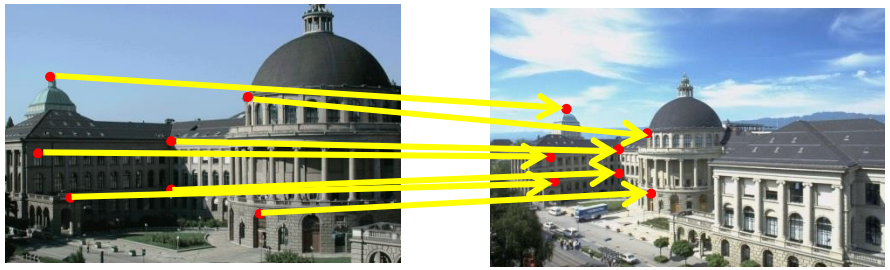
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

2D-to-2D
Motion from Image Feature Correspondences

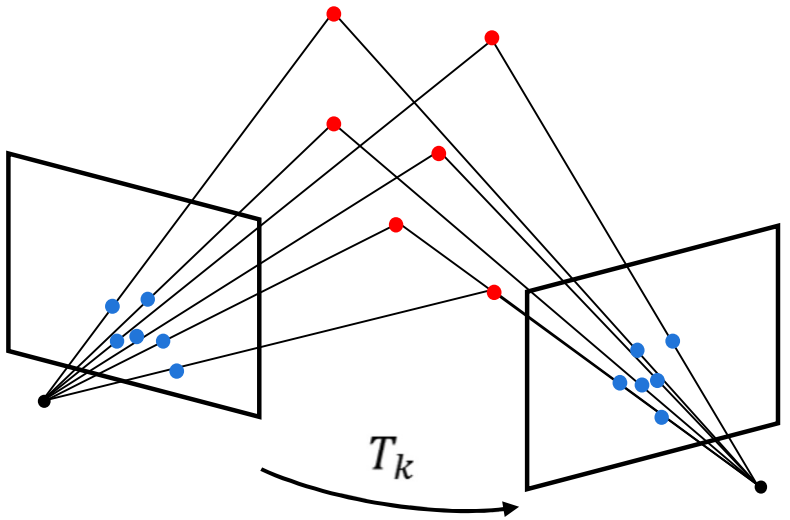
Motion estimation		
2D-2D	3D-3D	3D-2D

- Both f_{k-1} and f_k are specified in 2D
- The minimal-case solution involves 5-point correspondences
- The solution is found by determining the transformation that minimizes the reprojection error of the triangulated points in each image

$$T_k = \begin{bmatrix} R_{k,k-1} & t_{k,k-1} \\ 0 & 1 \end{bmatrix} = \arg\min_{X^i, C_k} \sum_{i,k} \| p_k^i - g(X^i, C_k) \|^2$$



I_{k-1} I_k



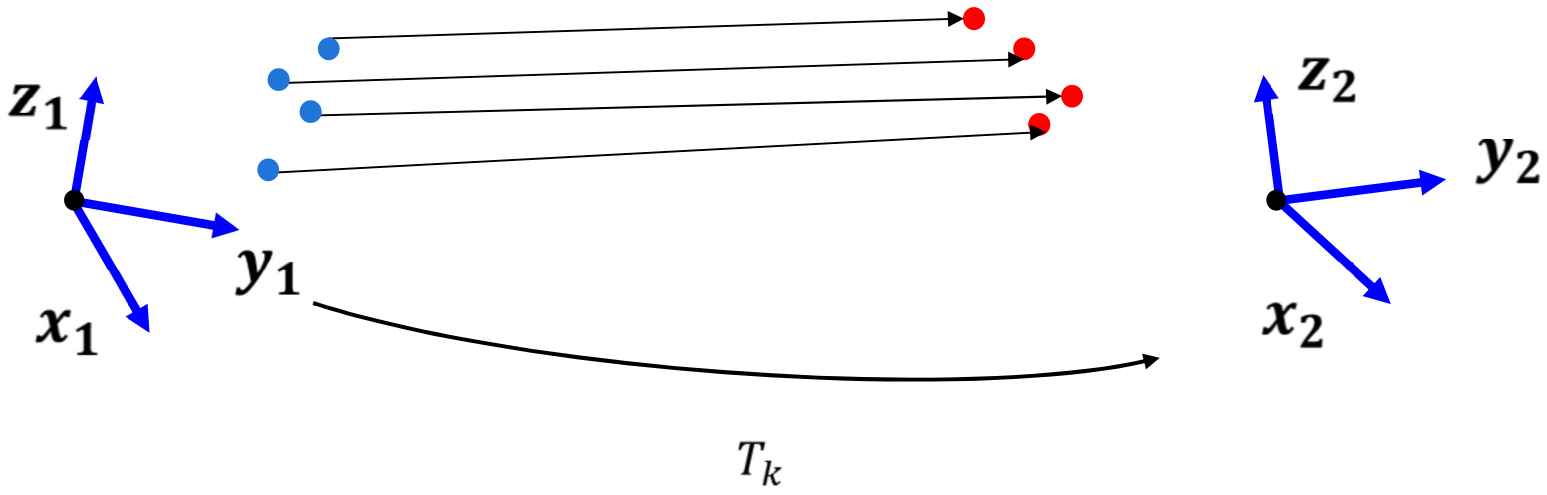
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

3D-to-3D
Motion from 3D-3D Point Correspondences

Motion estimation		
2D-2D	3D-3D	3D-2D

- Both f_{k-1} and f_k are specified in 3D
- To do this, it is necessary to triangulate 3D points (e.g. use a stereo camera)
- The minimal-case solution involves 3 non-collinear correspondences
- The solution is found by determining the aligning transformation that minimizes the 3D-3D distance

$$T_k = \begin{bmatrix} R_{k,k-1} & t_{k,k-1} \\ 0 & 1 \end{bmatrix} = \arg\min_{T_k} \sum_i ||\tilde{X}_k^i - T_k \tilde{X}_{k-1}^i||$$



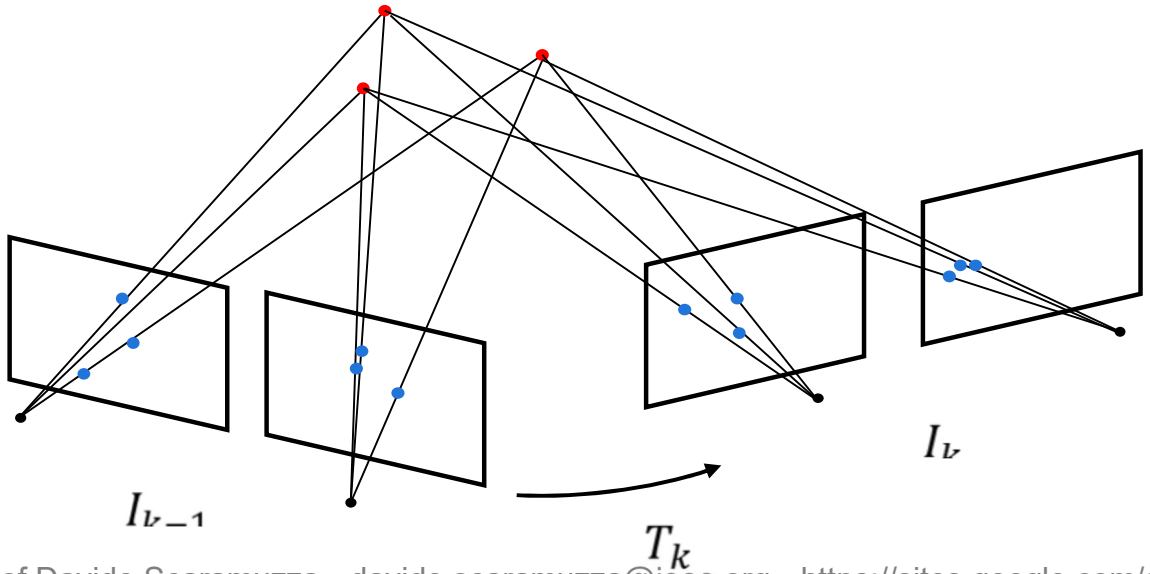
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

3D-to-3D
Motion from 3D-3D Point Correspondences

Motion estimation		
2D-2D	3D-3D	3D-2D

- Both f_{k-1} and f_k are specified in 3D
- To do this, it is necessary to triangulate 3D points (e.g. use a stereo camera)
- The minimal-case solution involves 3 non-collinear correspondences
- The solution is found by determining the aligning transformation that minimizes the 3D-3D distance

$$T_k = \begin{bmatrix} R_{k,k-1} & t_{k,k-1} \\ 0 & 1 \end{bmatrix} = \arg\min_{T_k} \sum_i ||\tilde{X}_k^i - T_k \tilde{X}_{k-1}^i||$$

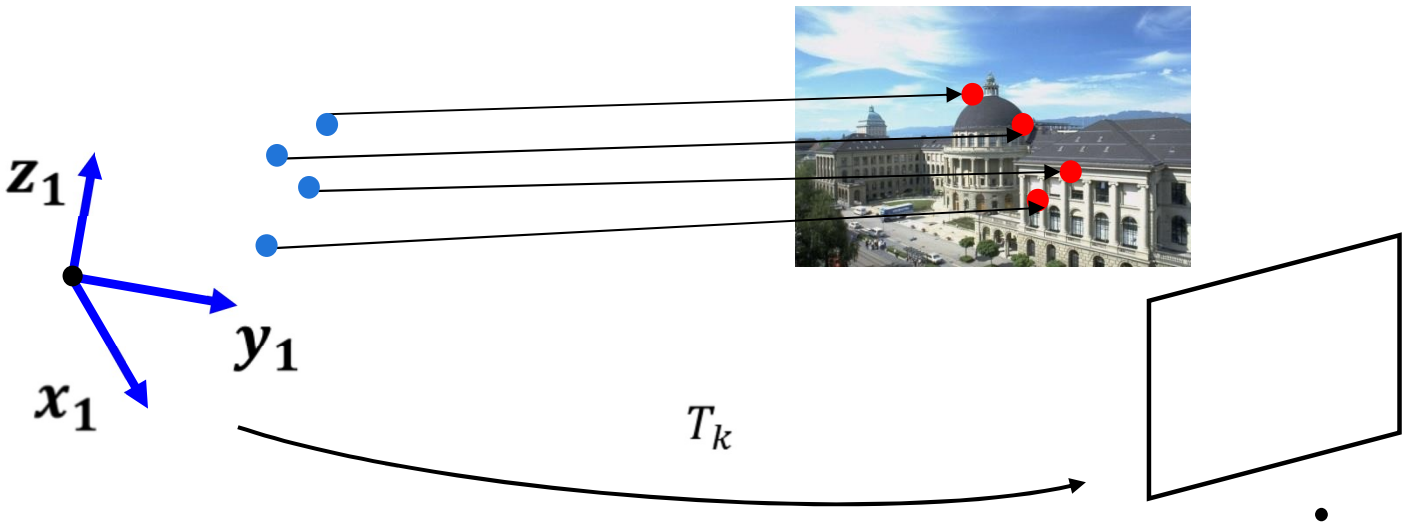


Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

3D-to-2D
Motion from 3D Structure and Image Correspondences

- f_{k-1} is specified in 3D and f_k in 2D
- This problem is known as *camera resection* or PnP (perspective from *n* points)
- The minimal-case solution involves 3 correspondences
- The solution is found by determining the transformation that minimizes the reprojection error

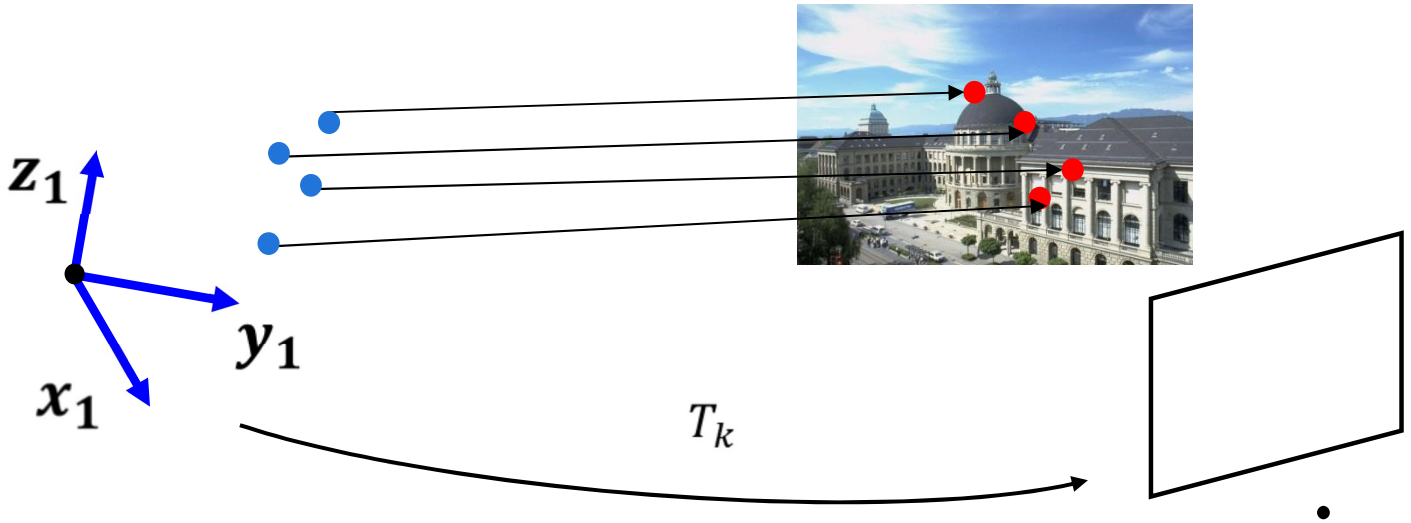
$$T_k = \begin{bmatrix} R_{k,k-1} & t_{k,k-1} \\ 0 & 1 \end{bmatrix} = \arg\min_{T_k} \sum_i \| p_k^i - \hat{p}_{k-1}^i \|^2$$



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

3D-to-2D
Motion from 3D Structure and Image Correspondences

- In the monocular case, the 3D structure needs to be triangulated from two adjacent camera views (e.g., I_{k-2} and I_{k-1}) and then matched to 2D image features in a third view (e.g., I_k).



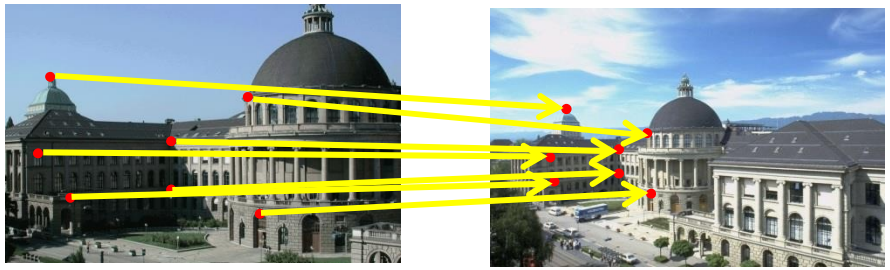
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

2D-to-2D
Motion from Image Feature Correspondences

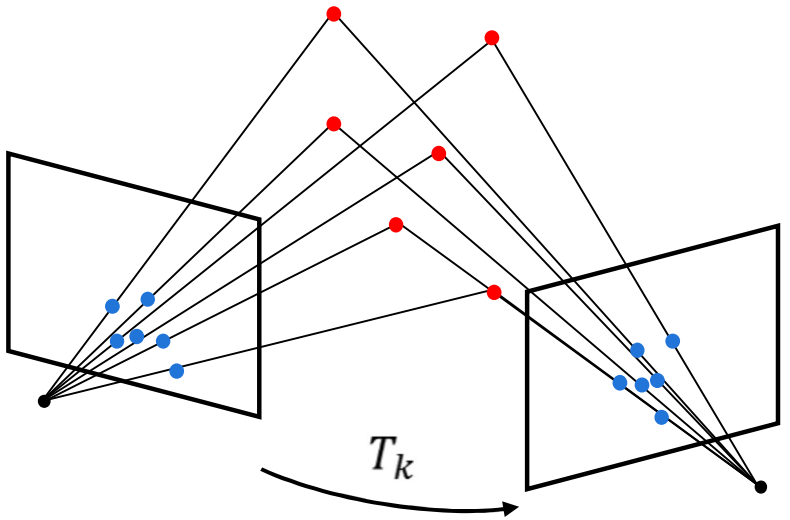
Motion estimation		
2D-2D	3D-3D	3D-2D

- Both f_{k-1} and f_k are specified in 2D
- The minimal-case solution involves 5-point correspondences
- The solution is found by determining the transformation that minimizes the reprojection error of the triangulated points in each image

$$T_k = \begin{bmatrix} R_{k,k-1} & t_{k,k-1} \\ 0 & 1 \end{bmatrix} = \arg\min_{X^i, C_k} \sum_{i,k} \| p_k^i - g(X^i, C_k) \|^2$$



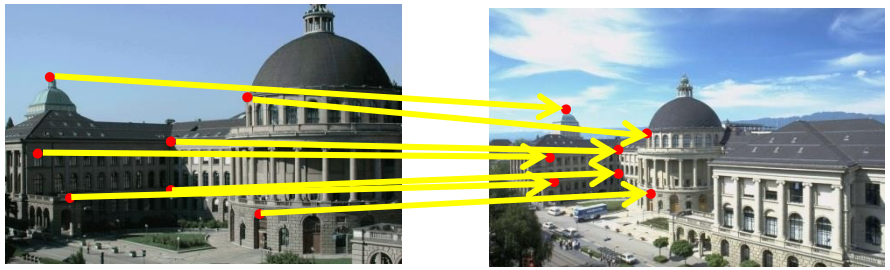
I_{k-1} I_k



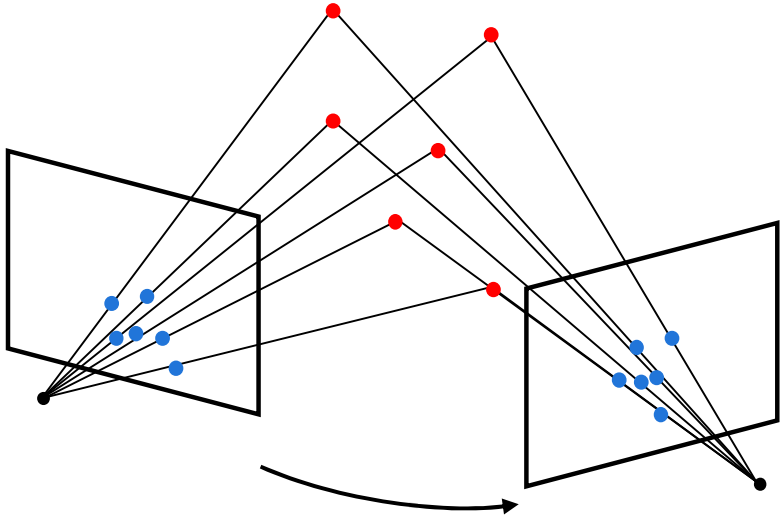
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

2D-to-2D
Motion from Image Feature Correspondences

Motion estimation		
2D-2D	3D-3D	3D-2D



I_{k-1} I_k

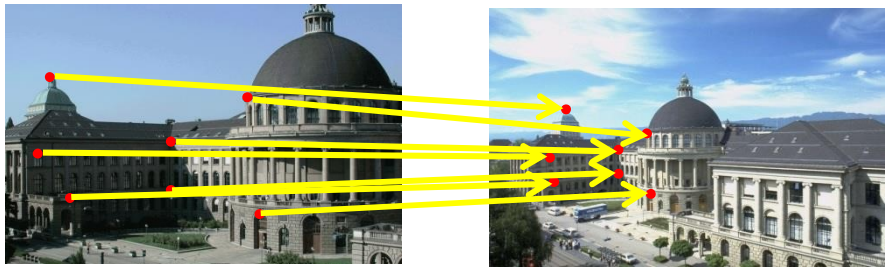


$$T_k = \begin{bmatrix} R_{k,k-1} & t_{k,k-1} \\ 0 & 1 \end{bmatrix} = ?$$

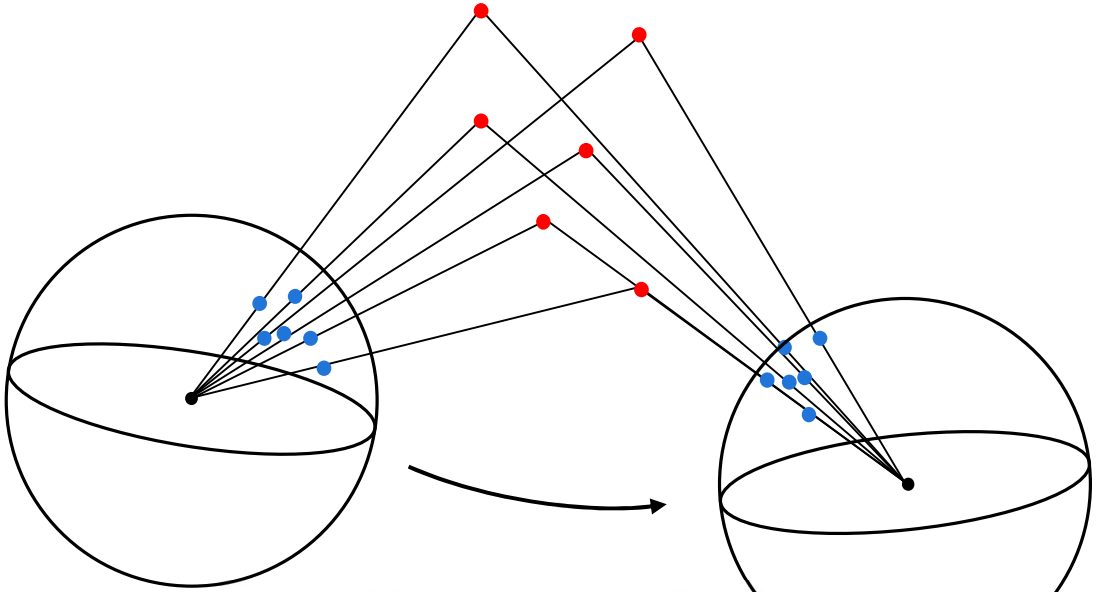
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

2D-to-2D
Motion from Image Feature Correspondences

Motion estimation		
2D-2D	3D-3D	3D-2D



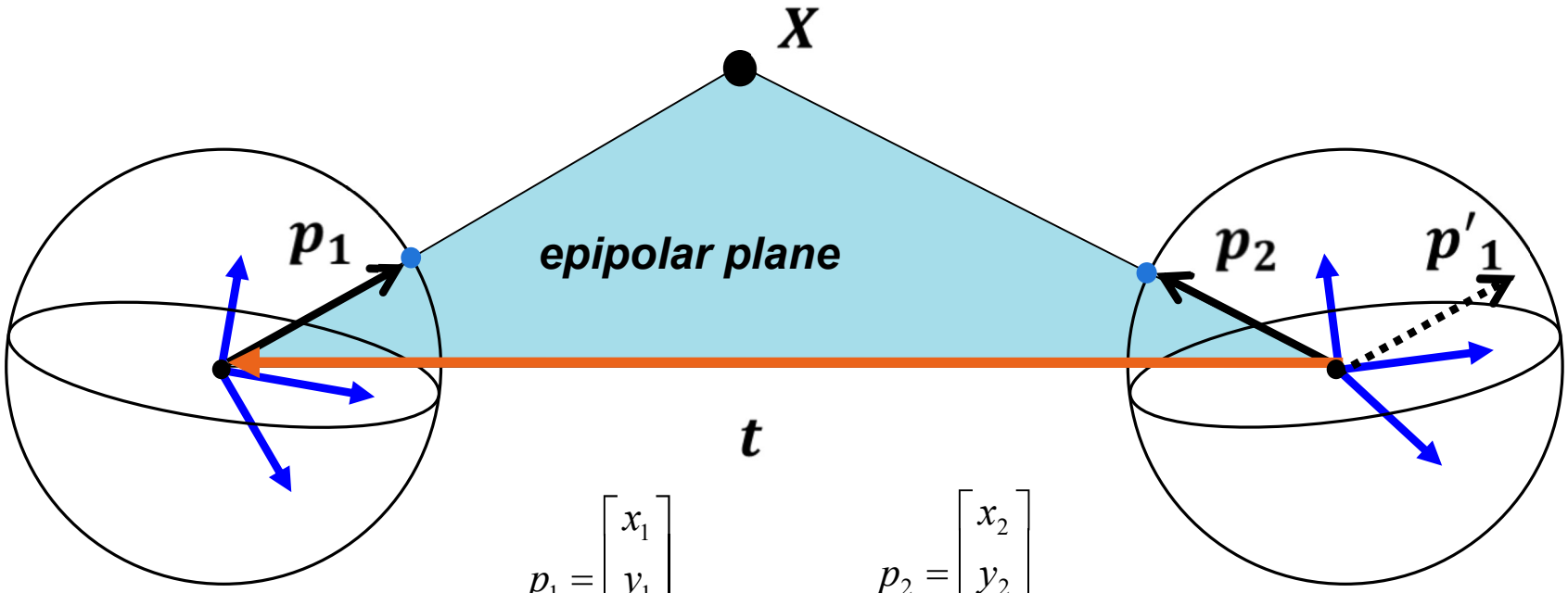
I_{k-1} I_k



$$T_k = \begin{bmatrix} R_{k,k-1} & t_{k,k-1} \\ 0 & 1 \end{bmatrix} = ?$$

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

2D-to-2D
Epipolar Geometry



$$p_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \qquad p_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

p_1, p_2, T are coplanar:

$$p_2^T \cdot (t \times p_1') = 0 \qquad \Rightarrow p_2^T \cdot (t \times (R p_1)) = 0$$

$$\Rightarrow p_2^T [t]_\times R \, p_1 = 0 \qquad \Rightarrow \boxed{p_2^T \, E \, p_1 = 0} \quad \textit{Epipolar constraint}$$

$$\boxed{\mathrm{E} = [t]_\times R} \quad \textit{essential matrix}$$

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

2D-to-2D
Epipolar Geometry

$$p_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \quad p_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$ ***Image coordinates on the Unit sphere***

$$p_2^T \, E \, p_1 = 0$$ ***Epipolar constraint***

$$\mathrm{E} = [t]_\times R$$ ***Essential matrix***

$$[\boldsymbol{t}]_\times = \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix}$$

- The Essential Matrix can be computed directly from the image coordinates (using SVD).
- At least 5 points needed! [Kruppa, 1913]. The more points, the better!
- The Essential Matrix can be decomposed into $\boldsymbol{R}$ and $\boldsymbol{t}$ (again using SVD)

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

2D-to-2D
Computing the Essential Matrix

Motion estimation		
2D-2D	3D-3D	3D-2D

- The Essential matrix can be computed from 5 point correspondences using [Nister'2003] algorithm (5-*point algorithm)*
- *The 5-p algorithm* has become the standard for 2D-to-2D motion estimation, however, its implementation is not straightforward
- A simple and straightforward solution for $n \geq 8$ noncoplanar points is the Longuet-Higgins' *8-p algorithm*, which is summarized here:
- Let $p_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $p_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ be the coordinates one feature correspondence
- $E = \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} = \begin{bmatrix} e_{11} \\ \vdots \\ e_{33} \end{bmatrix}$
- ${p_2}^{\mathrm{T}} E p_1 = 0 \Rightarrow [\boldsymbol{x_1x_2}\ \boldsymbol{y_1x_2}\ \boldsymbol{z_1x_2}\ \boldsymbol{x_1y_2}\ \boldsymbol{y_1y_2}\ \ \boldsymbol{z_1y_2}\ \ \boldsymbol{x_1z_2}\ \boldsymbol{y_1z_2}\ \ \boldsymbol{z_1z_2}]E = 0$

  which can be solved with SVD

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

2D-to-2D Algorithm

Algorithm 1: VO from 2D-to-2D correspondeces

1 Capture new frame I_k
2 Extract and match features between I_{k-1} and I_k,
3 Compute essential matrix for image pair I_{k-1}, I_k
4 Decompose essential matrix into R_k and t_k, and form T_k
5 Compute relative scale and rescale t_k accordingly
6 Concatenate transformation by computing $C_k = C_{k-1}T_k$
7 Repeat from 1

How do we compute the relative scale between I_{k-2}, I_{k-1}, and I_k ?

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

2D-to-2D
Relative Scale Computation

- The absolute scale of the translation cannot be computed from two images.
- However, it is possible to compute relative scales for the subsequent transformations.
- One way of doing this is to triangulate 3D points X_{k-1} and X_k from two subsequent image pairs. From corresponding 3D points, the relative distances between any combination of two 3D points can be computed.
- The proper scale can then be determined from the distance ratio *r* between a point pair in X_{k-1} and a pair in X_k

$$r = \frac{||X_{k-1,i} - X_{k-1,j}||}{||X_{k,i} - X_{k,j}||}$$

- For robustness, the scale ratios for many point pairs are computed and the mean (or in presence of outliers the median) is used

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

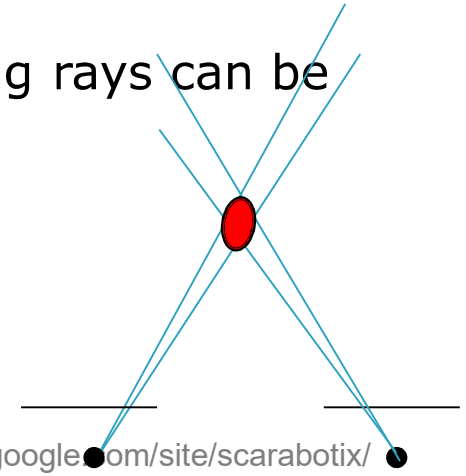
Motion Estimation: Summary

Type of correspondences	Monocular	Stereo
2D-2D	X	X
3D-3D		X
3D-2D	X	X

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Triangulation and Keyframe Selection

- Some of the previous motion estimation methods require triangulation of 3D points
- Triangulated 3D points are determined by intersecting backprojected rays from 2D image correspondences of at least two image frames
- In reality, they never intersect due to
  - image noise,
  - camera model and calibration errors,
  - and feature matching uncertainty
- The point at minimal distance from all intersecting rays can be taken as an estimate of the 3D point position



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Triangulation and Keyframe Selection

- When frames are taken at nearby positions compared to the scene distance, 3D points will exibit large uncertainty

- **Therefore, 3D-3D motion estimation methods will drift much more quickly than 3D-2D and 2D-2D methods**

- In fact, the uncertainty introduced by triangulation affects the motion estimation. In fact, in the 3D-to-3D case the 3D position error is minimized, while in the 3D-to-2D and 2D-to-2D cases is the image reprojection error

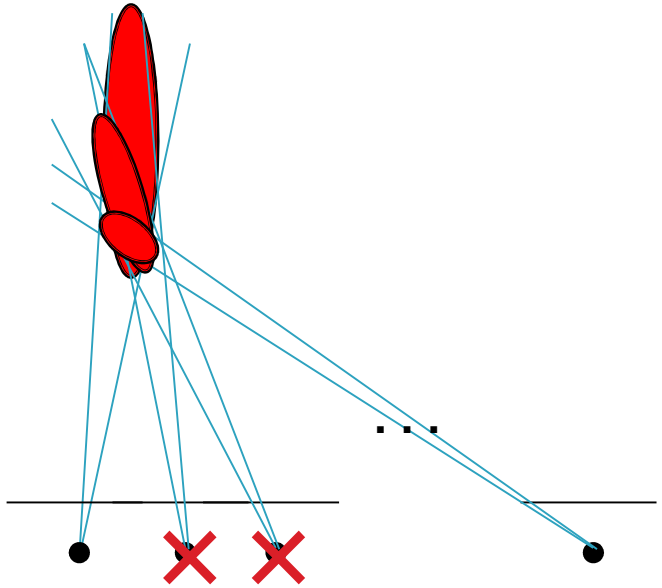


Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Triangulation and Keyframe Selection

- One way to avoid this consists of skipping frames until the average uncertainty of the 3D points decreases below a certain threshold. The selected frames are called *keyframes*

- Keyframe selection is a very important step in VO and should always be done before updating the motion



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

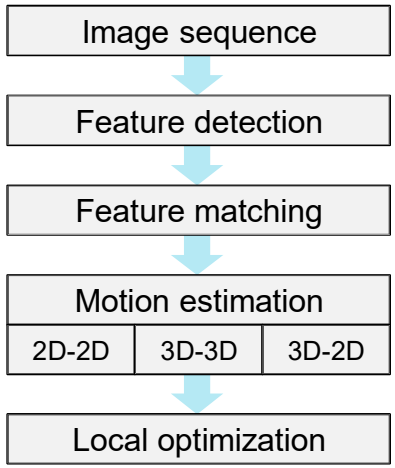
Summary Considerations

- In the Stereo vision case, 3D-2D method exhibits less drift than 3D-3D method
- Stereo vision has the advantage over monocular vision that both motion and structure are computed in the absolute scale. It also exhibits less drift.
- When the distance to the scene is much larger than the stereo baseline, stereo VO degenerates into monocular VO
- Keyframes should be selected carefully to reduce drift
- Regardless of the chosen motion computation method, local bundle adjustment (over the last *m* frames) should be always performed to compute a more accurate estimate of the trajectory. **After bundle adjustment, the effects of the motion estimation method are much more alleviated** (as long as the initialization is close to the solution)

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Outline

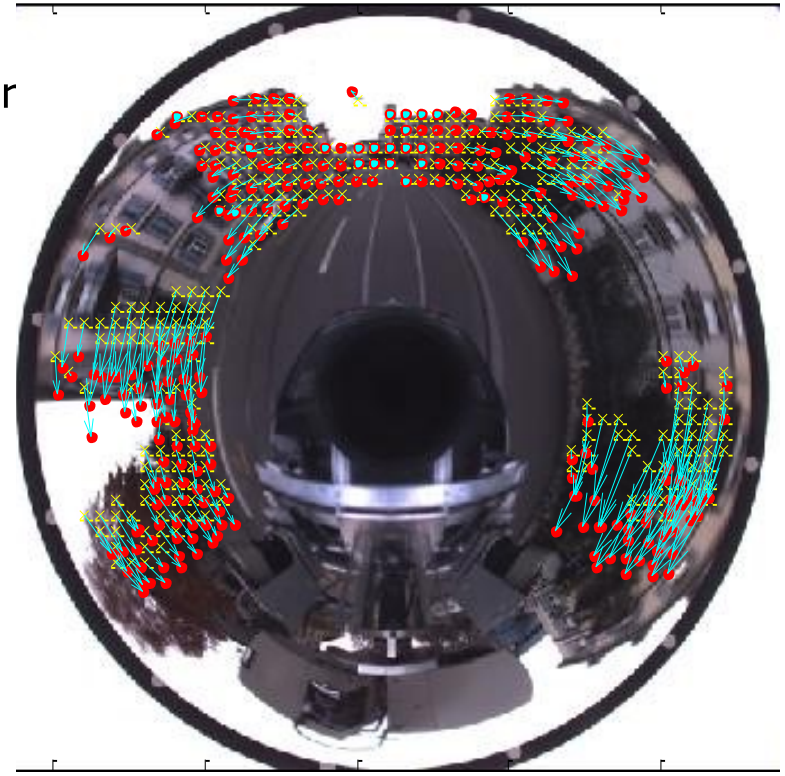
- Brief history of VO
- Problem formulation
- Camera modeling and calibration
- Motion estimation
- Robust estimation
- Error propagation
- Camera-pose optimization (bundle adjustment)
- Discussion



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

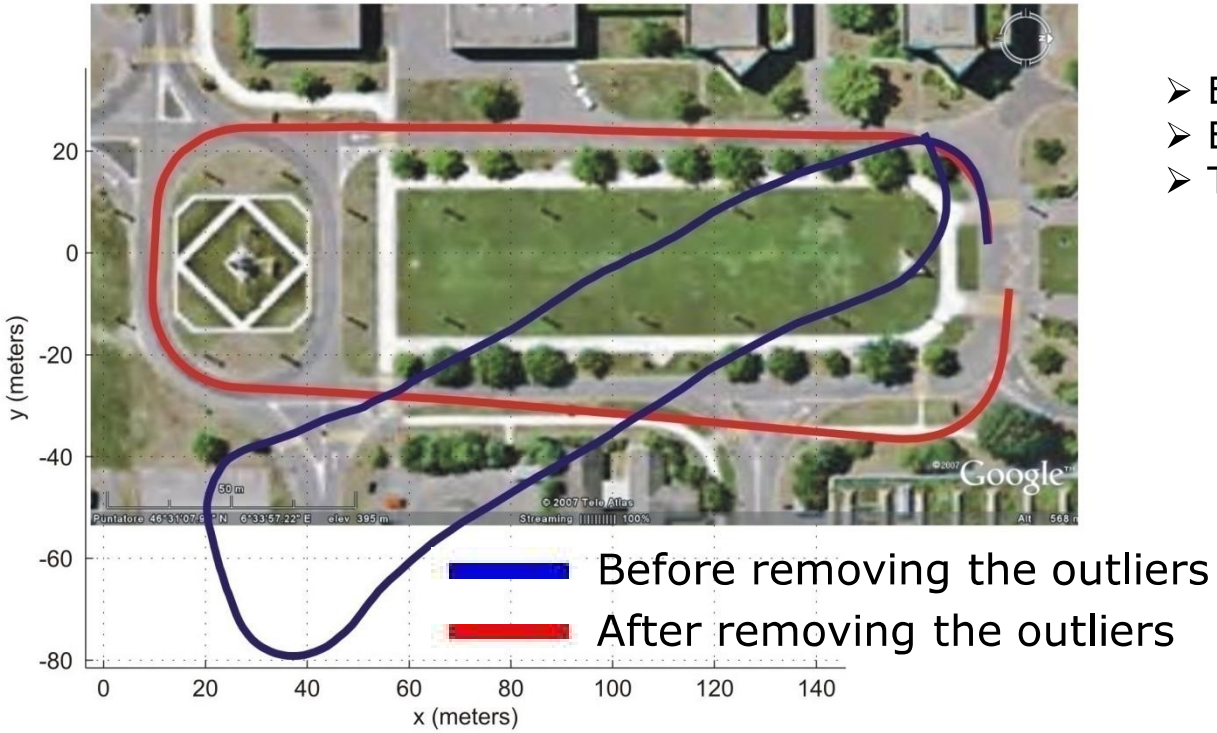
Robust Estimation

- Matched points are usually contaminated by outliers, that is, wrong data associations
- Possible causes of outliers are
  - image noise,
  - occlusions,
  - blur,
  - changes in view point and illumination for which the mathematical model of the feature detector or descriptor does not account for
- For the camera motion to be estimated accurately, outliers must be removed
- This is the task of Robust Estimation



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Influence of Outliers on Motion Estimation



- Error at the loop closure: 6.5 m
- Error in orientation: 5 deg
- Trajectory length: 400 m

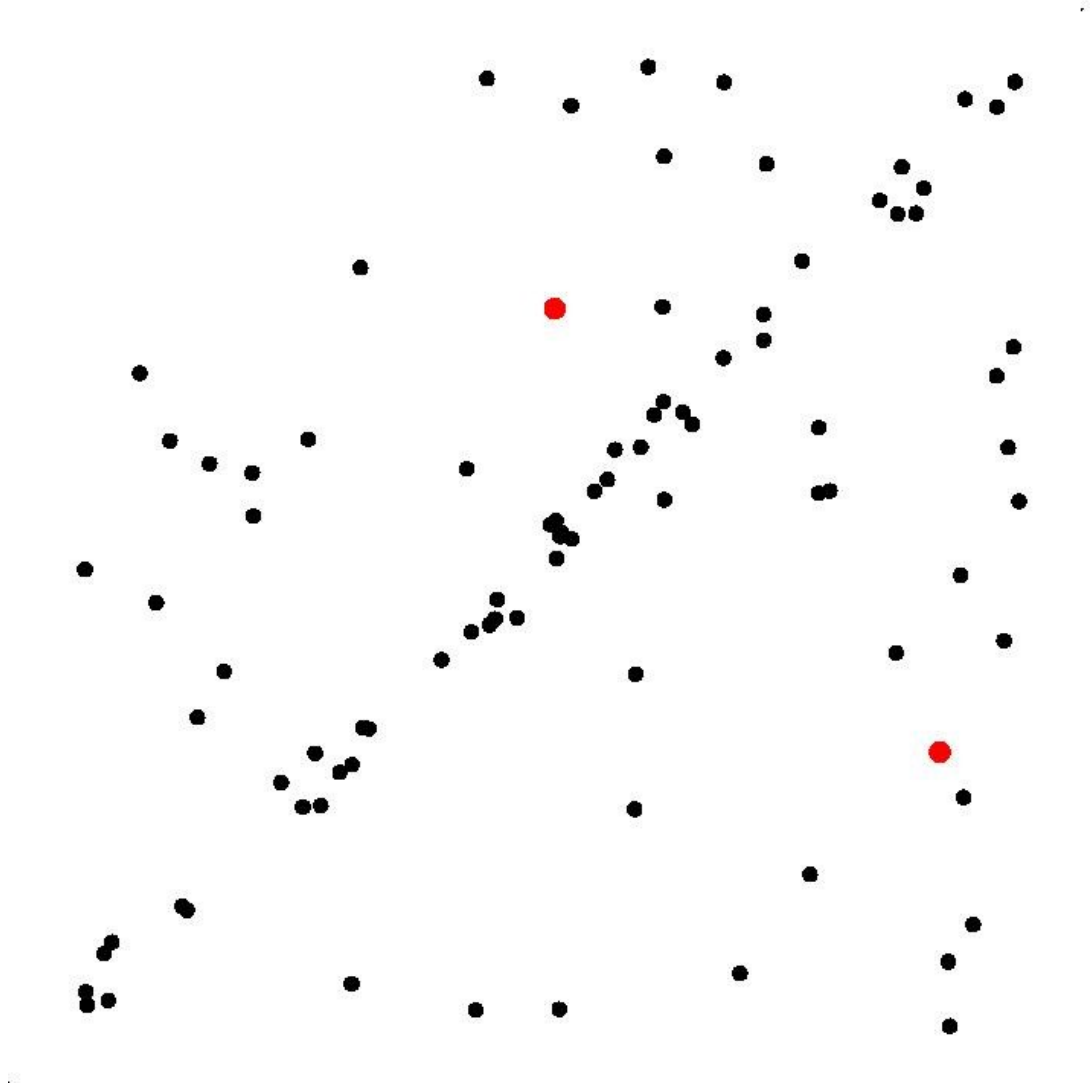
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

RANSAC Example: Line Extraction



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

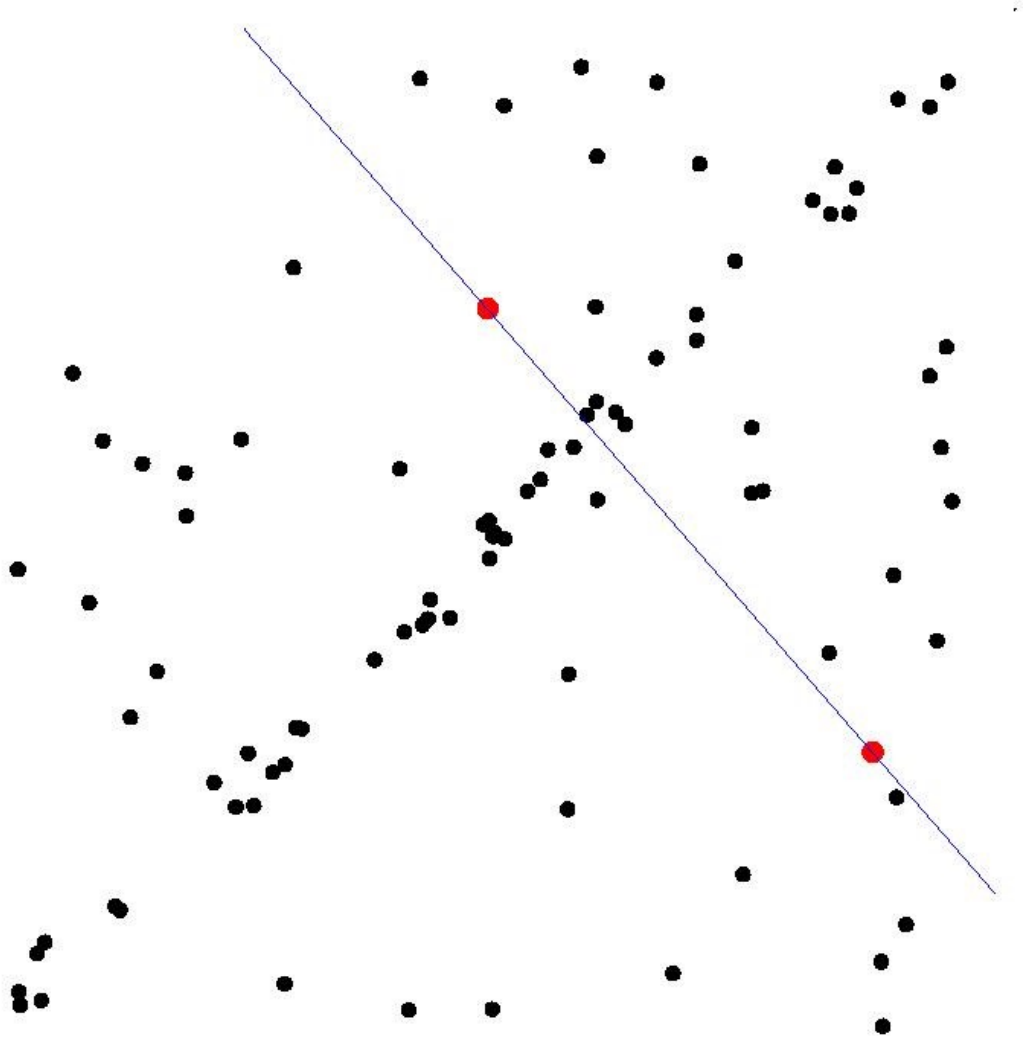
RANSAC Example: Line Extraction



- Select sample of 2 points at random

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

RANSAC Example: Line Extraction

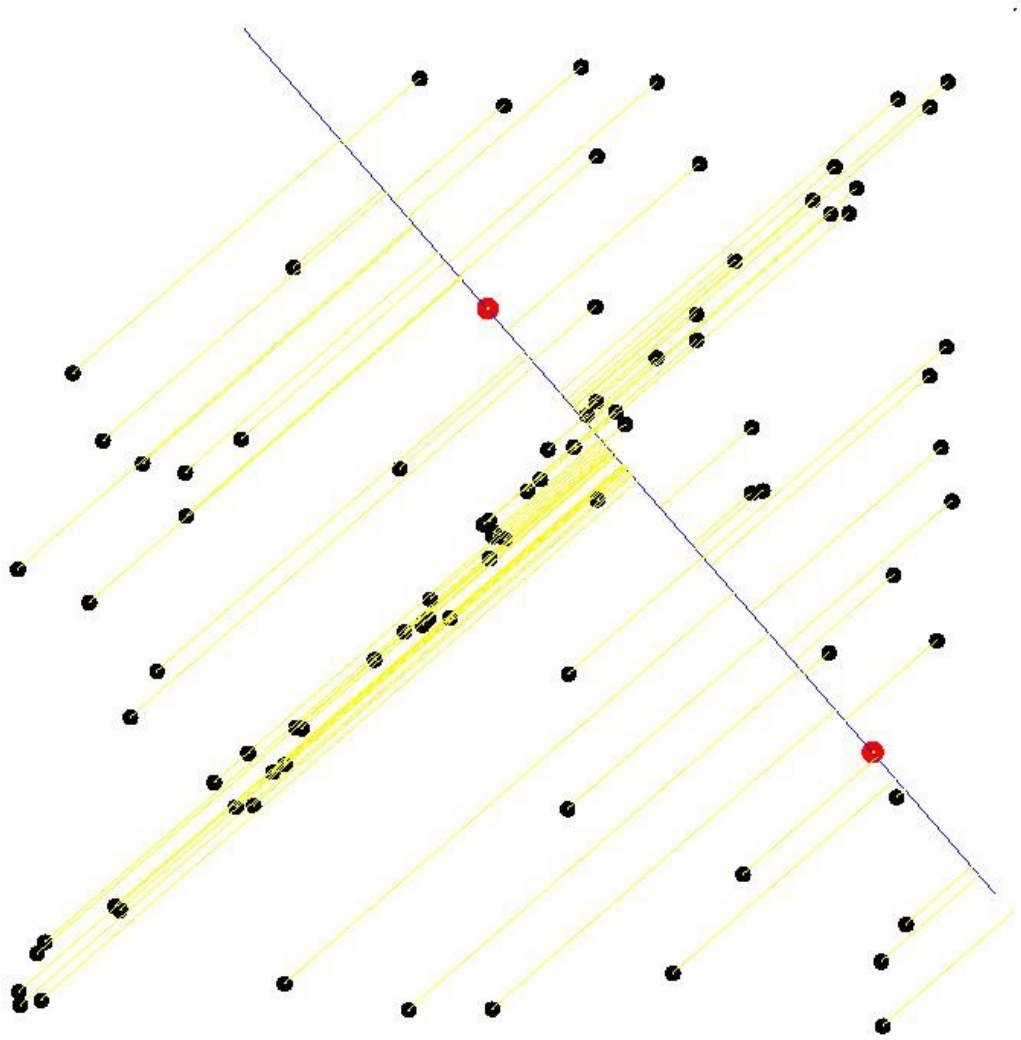


- Select sample of 2 points at random

- Calculate model parameters that fit the data in the sample

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

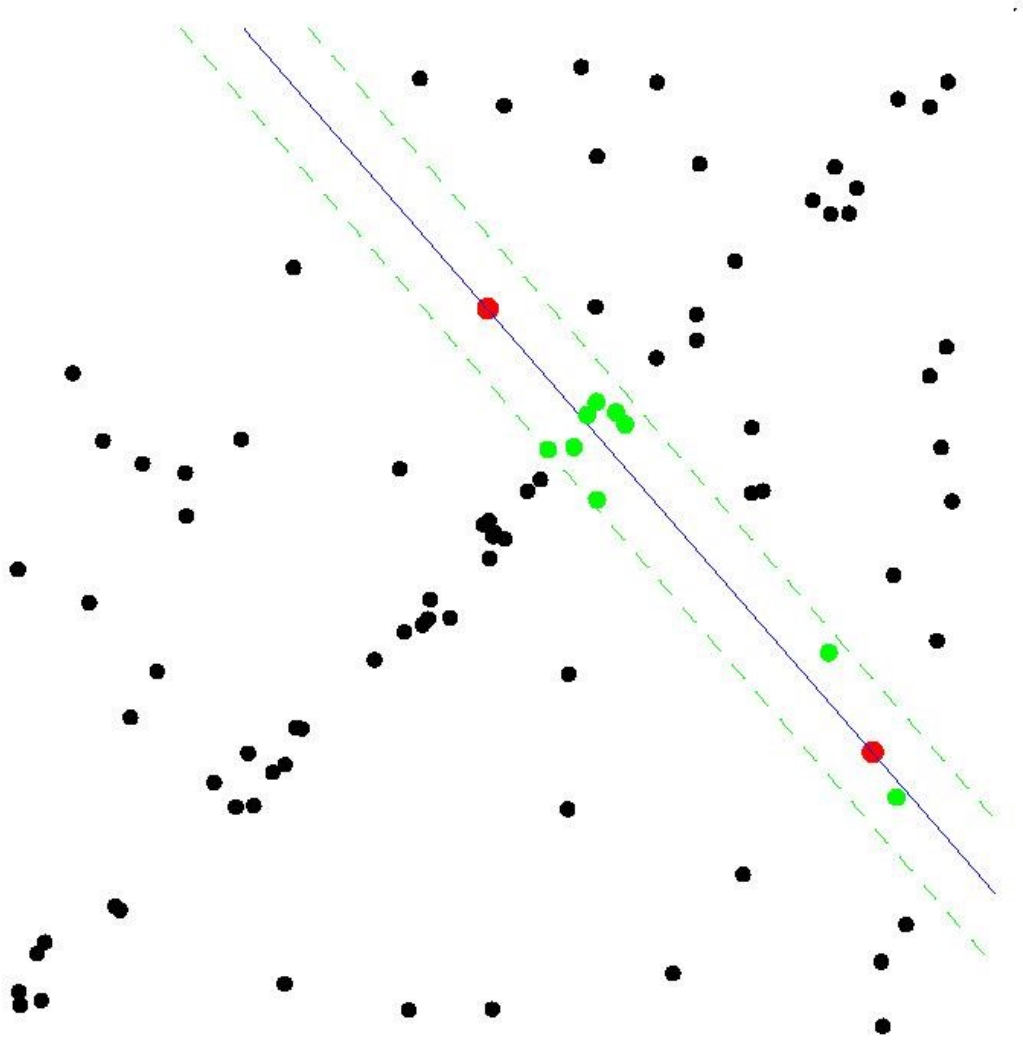
RANSAC Example: Line Extraction



- Select sample of 2 points at random
- Calculate model parameters that fit the data in the sample
- Calculate error function for each data point

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

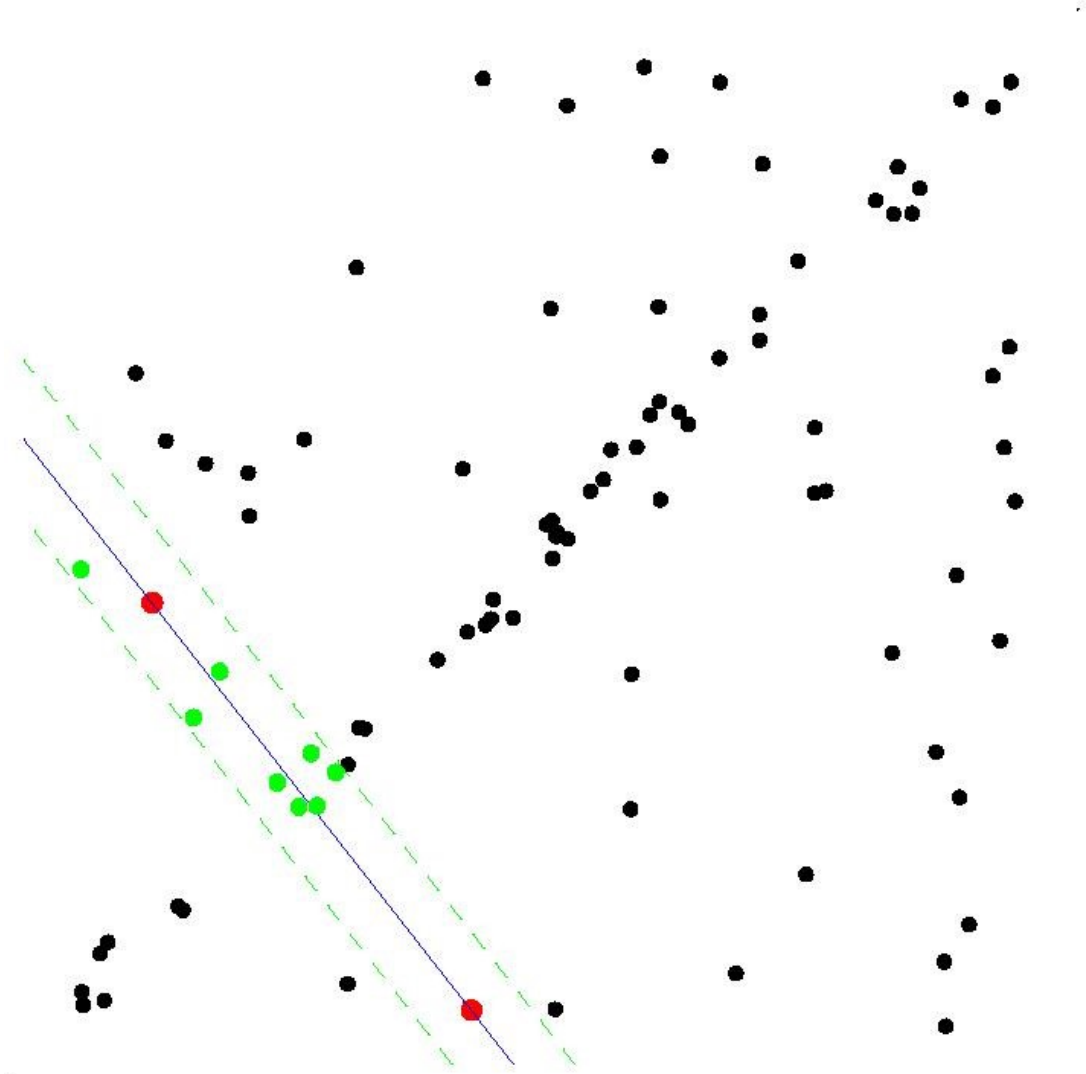
RANSAC Example: Line Extraction



- Select sample of 2 points at random
- Calculate model parameters that fit the data in the sample
- Calculate error function for each data point
- Select data that support current hypothesis

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

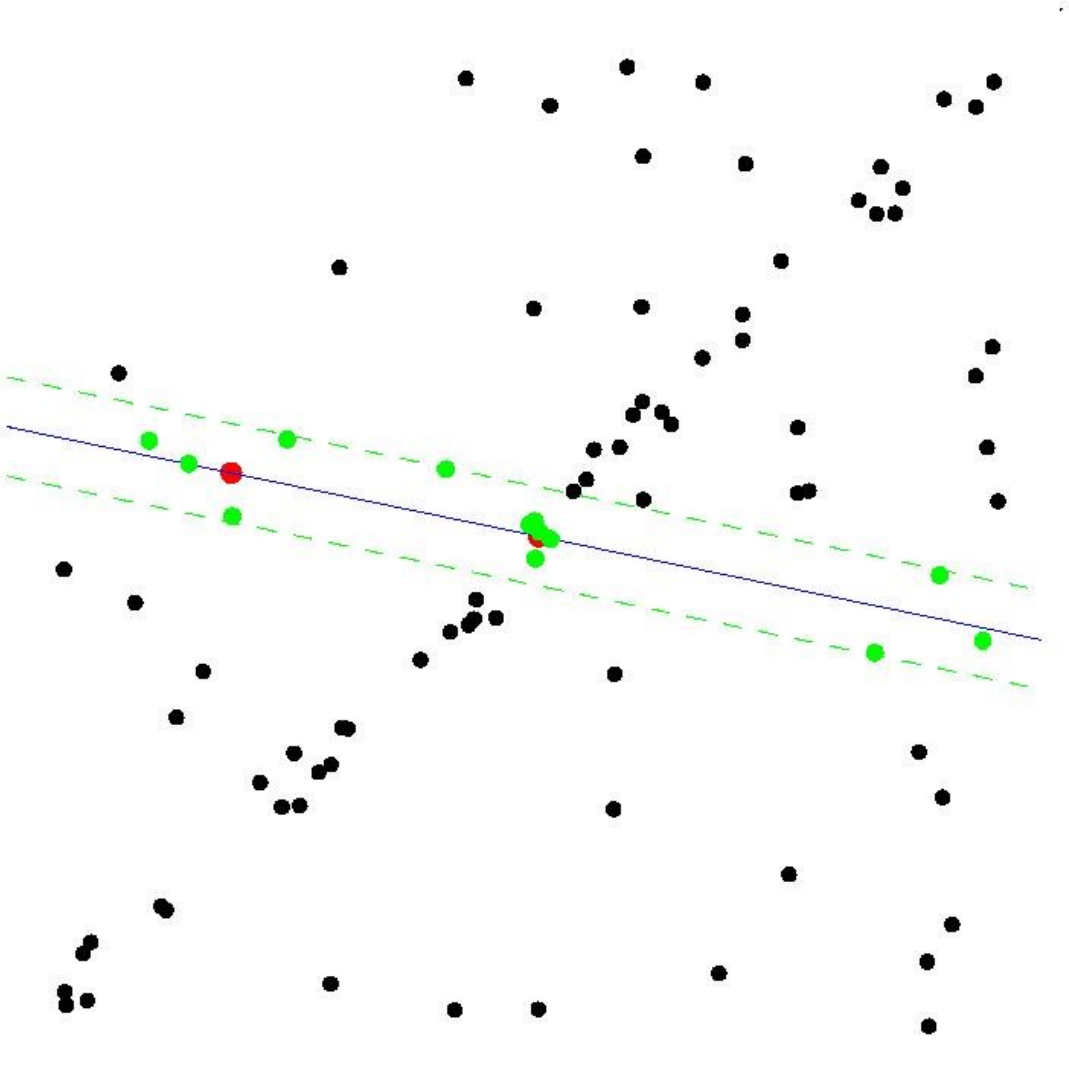
RANSAC Example: Line Extraction



- Select sample of 2 points at random
- Calculate model parameters that fit the data in the sample
- Calculate error function for each data point
- Select data that support current hypothesis
- Repeat sampling

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

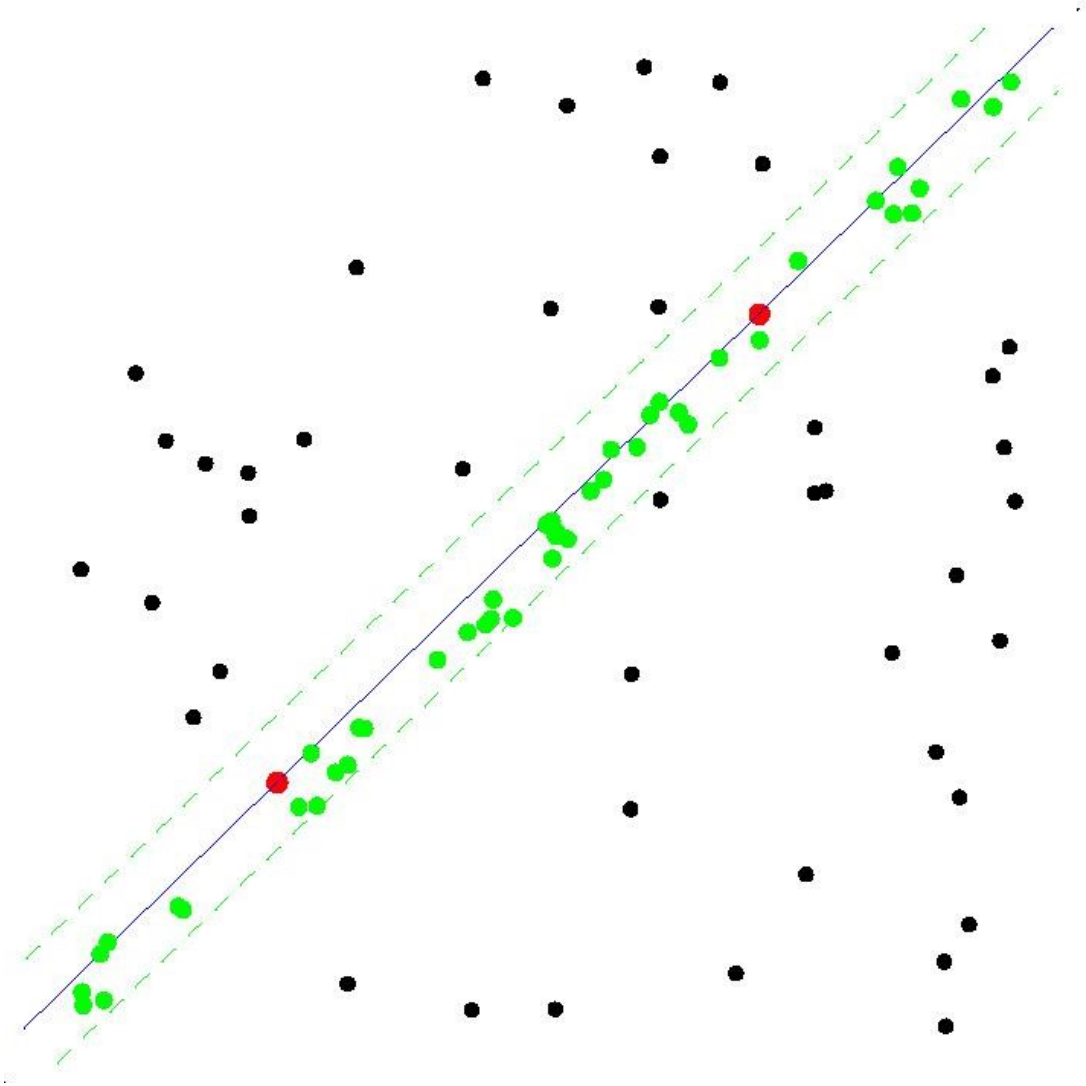
RANSAC Example: Line Extraction



- Select sample of 2 points at random
- Calculate model parameters that fit the data in the sample
- Calculate error function for each data point
- Select data that support current hypothesis
- Repeat sampling

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

RANSAC Example: Line Extraction

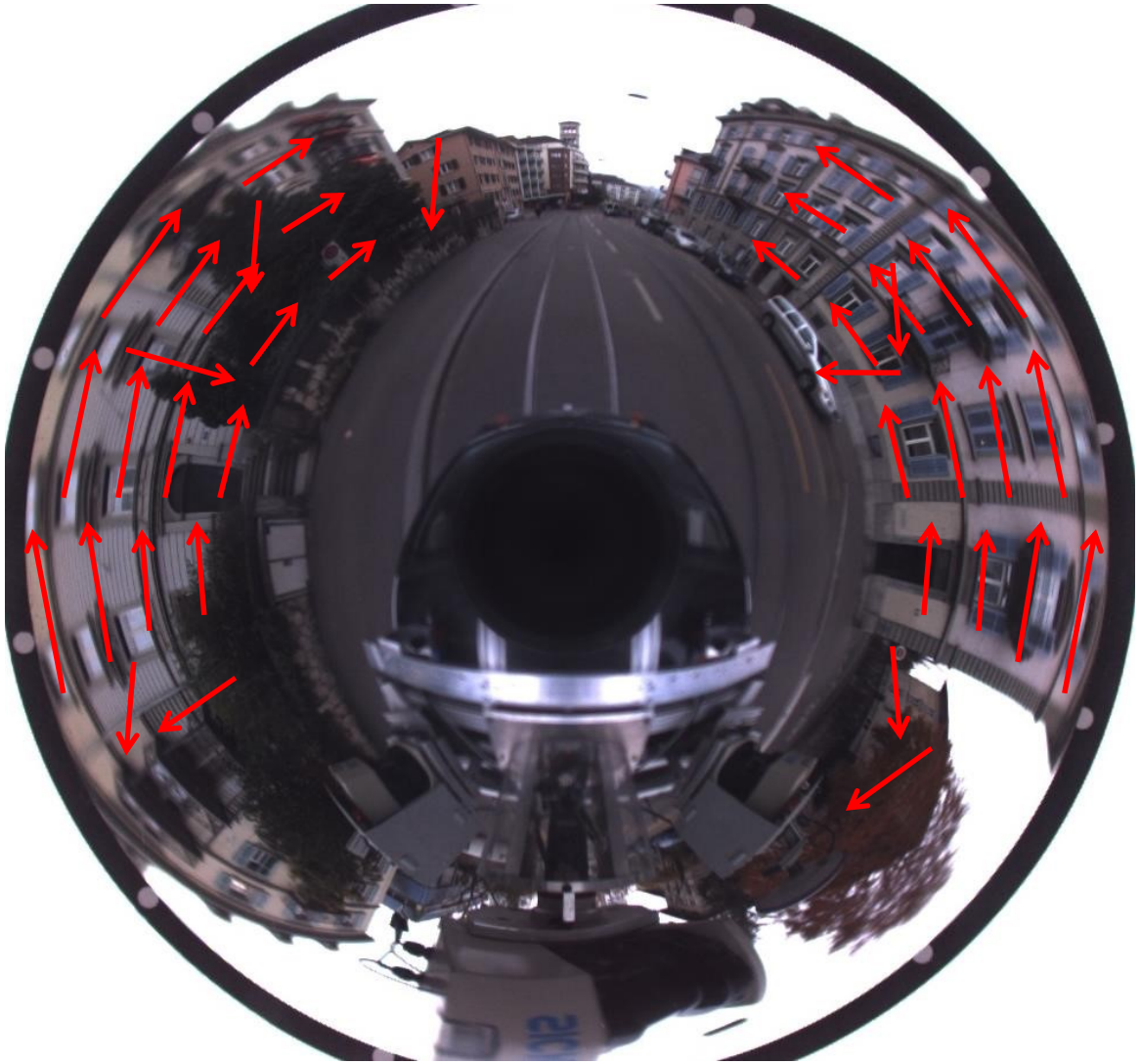


ALL-INLIER SAMPLE

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

RANSAC [Fishler & Bolles, 1981]

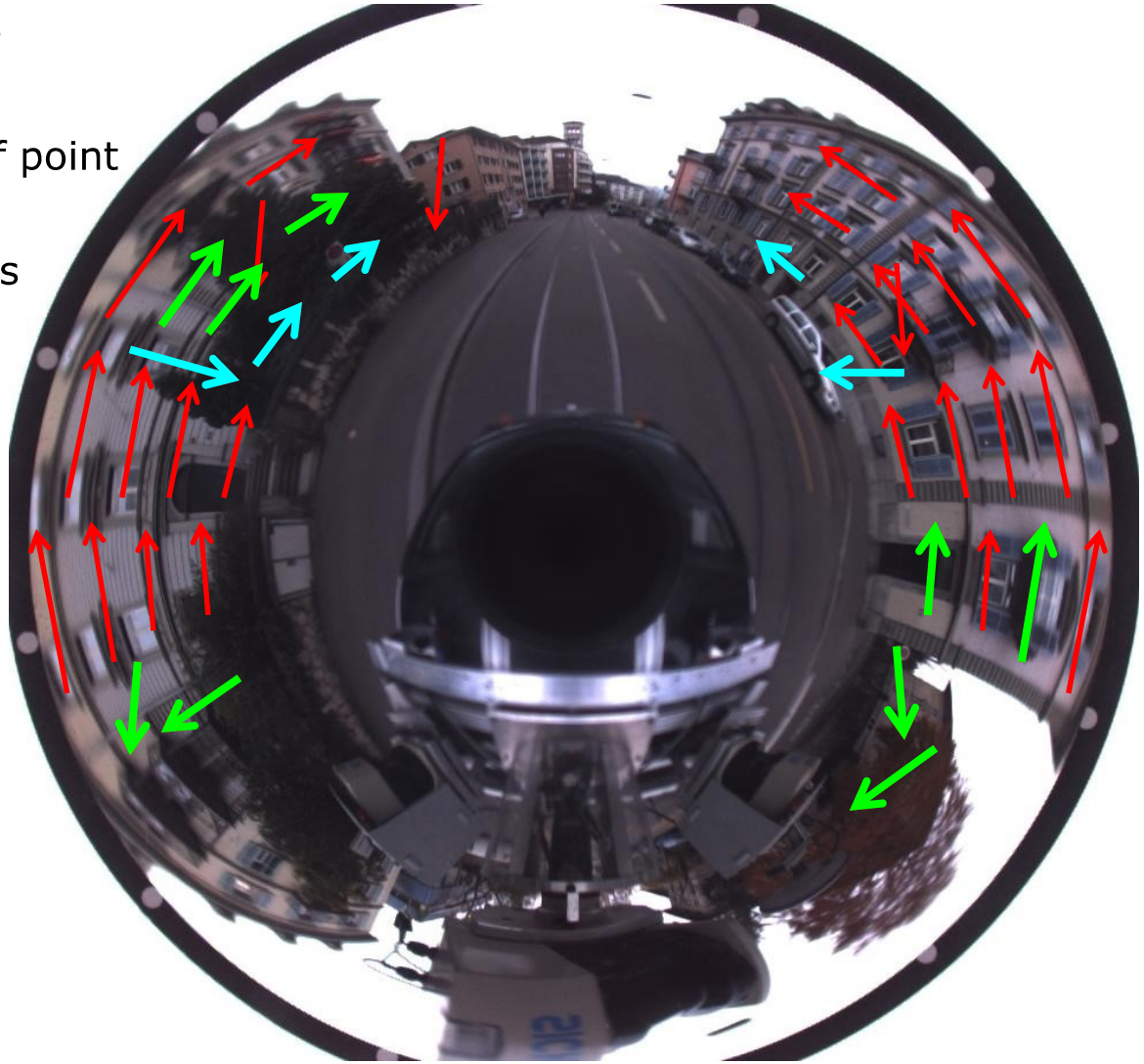
Has been established as the standard method for motion estimation in the presence of outliers



RANSAC [Fishler & Bolles, 1981]

Has been established as the standard method for motion estimation in the presence of outliers

1. Randomly select a minimal set of point correspondences
2. Compute motion and count inliers
3. Repeat from 1



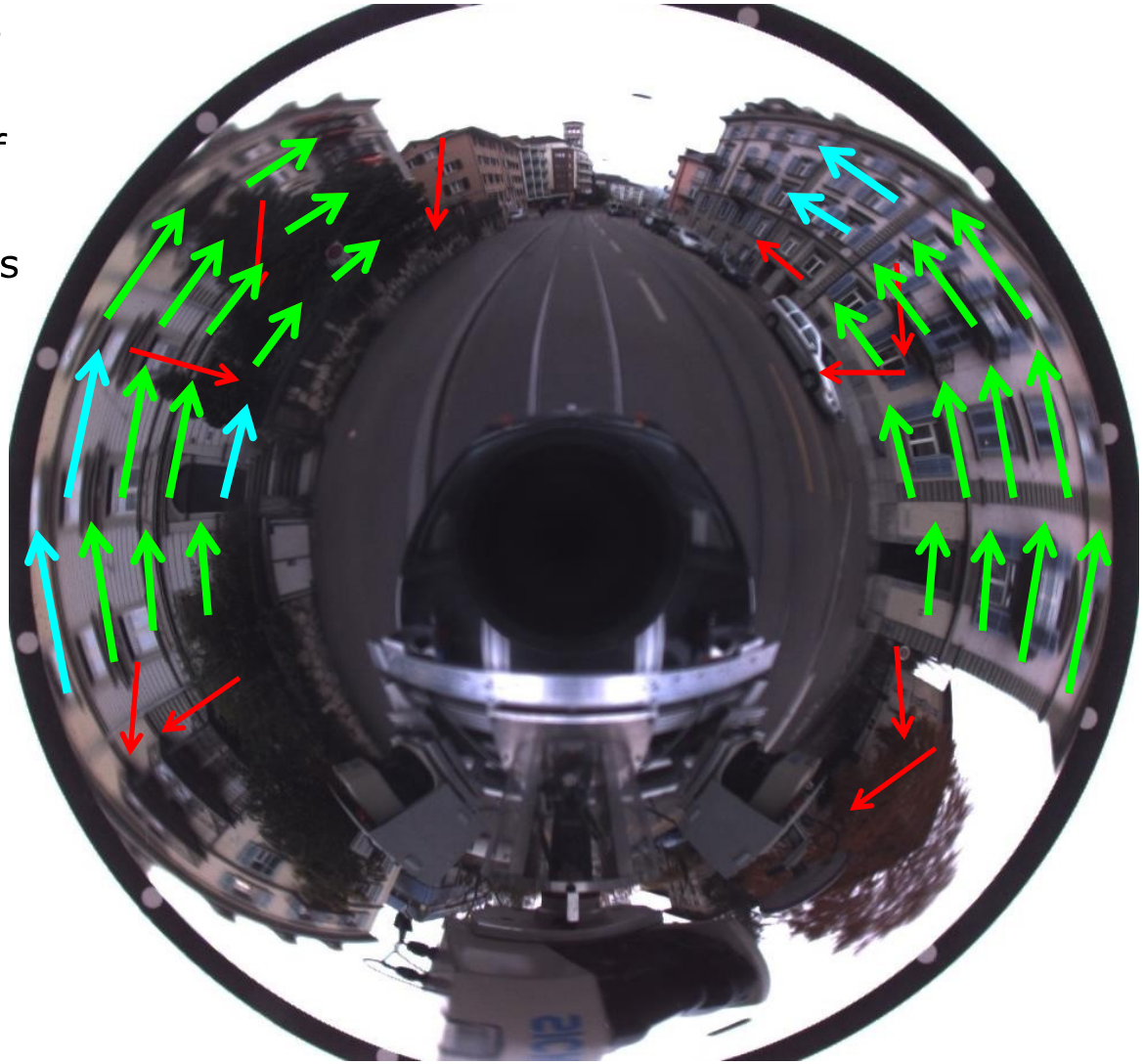
RANSAC [Fishler & Bolles, 1981]

Has been established as the standard method for motion estimation in the presence of outliers

1. Randomly select a minimal set of point correspondences
2. Compute motion and count inliers
3. Repeat N times

The number of iterations needed grows exponentially with the outliers

~ 1000 iterations!



RANSAC Algorithm

Algorithm 1: RANSAC

1) Initial: let A be a set of N feature correspondences
2) repeat
2.1) Randomly select a sample of s points from A
2.2) Fit a model to these points
2.3) Compute the distance of all other points to this model
2.4) Construct the inlier set (i.e. count the number of points whose distance from the model $< d$)
2.5) Store these inliers
2.6) until maximum number of iterations reached
3) The set with the maximum number of inliers is chosen as a solution to the problem
4) Estimate the model using all the inliers

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

How many iterations of RANSAC ?

- The number of iterations N that is necessary to guarantee that a correct solution is found can be computed by

$$N = \frac{\log(1-p)}{\log(1-(1-\varepsilon)^s)}$$

- $\boldsymbol{s}$ is the number of points from which the model can be instantiated
- $\boldsymbol{\varepsilon}$ is the percentage of outliers in the data
- $\boldsymbol{p}$ is the requested probability of success

- Example: $\boldsymbol{p = 99\%,\ s = 5,\ \varepsilon = 50\% \Rightarrow N = 145}$

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

How many iterations of RANSAC ?

- RANSAC is an iterative method and is **non deterministic** in that it **exhibits a different solution** on different runs; however, the solution tends to be stable when the number of iterations grows

- For the sake of robustness, in many practical implementations N is usually multiplied by a factor of 10

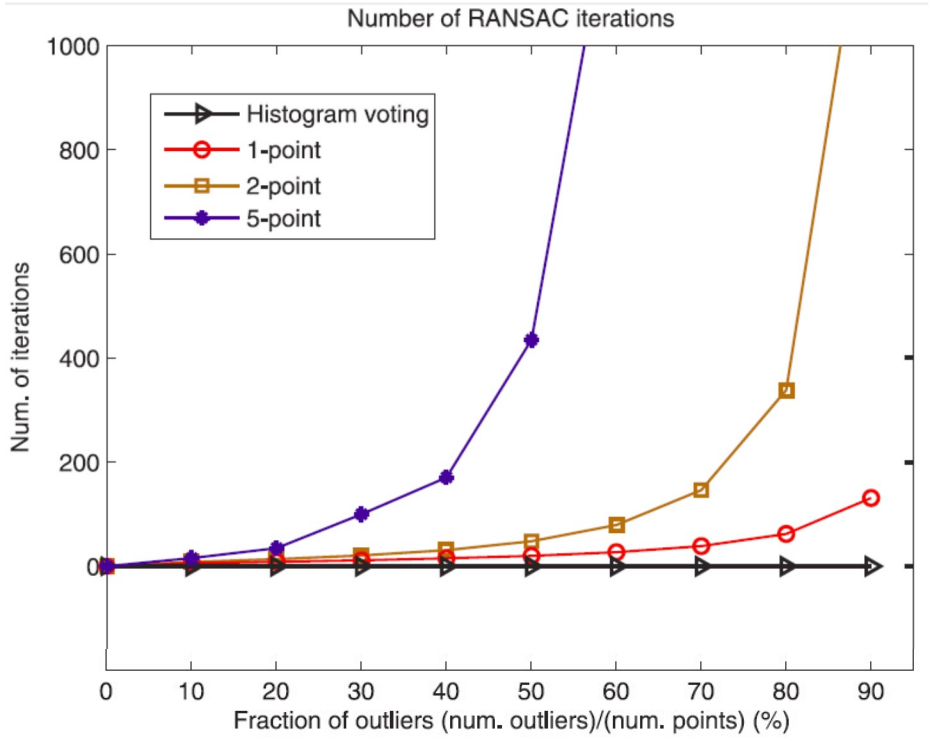
- More advanced implementations of RANSAC **estimate the fraction of inliers adaptively**, iteration after iteration (like in your exercise today)

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

RANSAC iterations

- N is exponential in the number of data points s necessary to estimate the model
- Therefore, there is a high interest in using a minimal parameterization of the model

Number of points (s):	8	7	6	5	4	2	1
Number of iterations (N):	1177	587	292	145	71	16	7



Copyright of I

What is the minimum number of points to estimate motion ?

To estimate the motion of a calibrated camera in 6 DoF,
we need 5 points
[Kruppa, 1913]

Why ?

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

What is the minimum number of points to estimate motion ?

In 6 DoF we would need 6 points ...

... but the scale is unobservable ...

... and therefore we only need 6 – 1 = 5 points

["5-Point RANSAC", Nister, 2003]

General rule:

Minimum number of points = N_{DoF} - 1

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

What is the minimum number of points to estimate motion ?

The “5-Point RANSAC” typically needs ~1000 iterations

To reduce the number iterations, we should use a smaller number of points (< 5)

Is this possible?

Yes, if we exploit motion constraints!

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

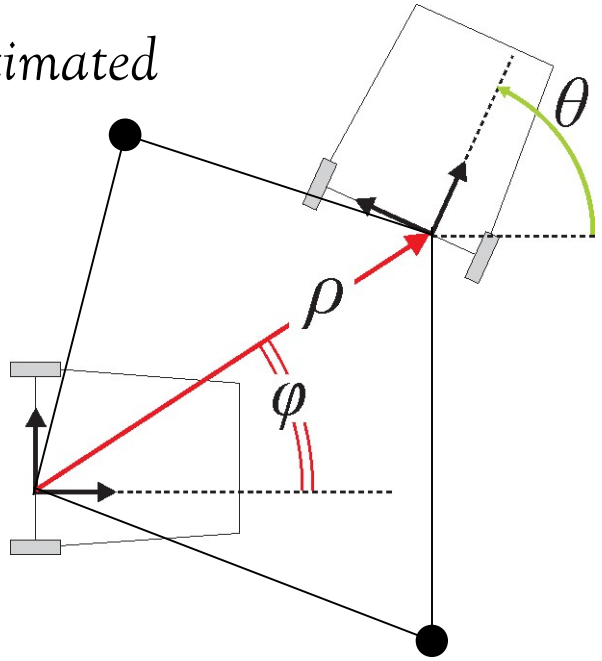
What is the minimum number of points to estimate motion ?

For planar motion, only 3 parameters need to be estimated

$\theta, \varphi, \rho \Rightarrow$ *3 DoF*

and therefore only 2 points are needed

["2-Point RANSAC", Ortin, 2001]



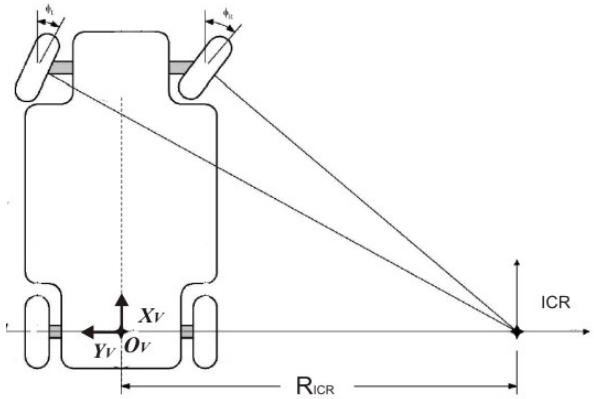
Can we use an even smaller number of points?

Yes, if we exploit the vehicle non-holonomic *constraints*

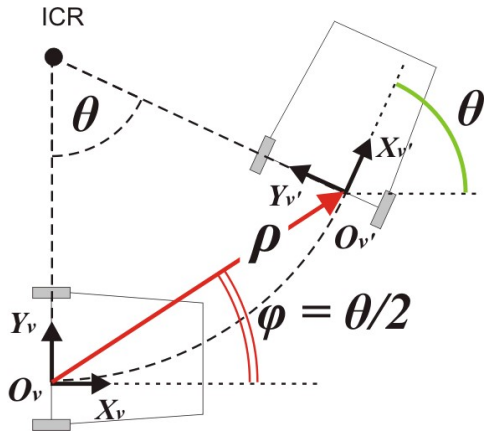
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Exploiting Vehicle's Non-Holonomic Constraints ?

- Wheeled vehicles follow locally circular motion about the Instantaneous Center of Rotation (ICR)



Example of Ackerman steering principle



Locally circular motion

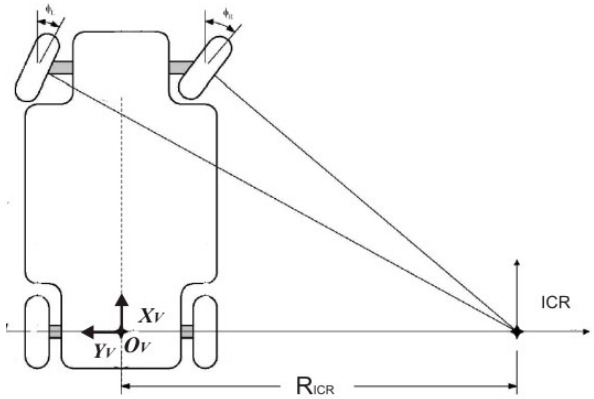




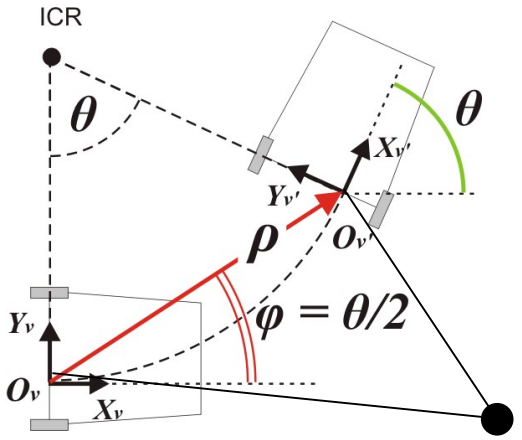
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Exploiting Vehicle's Non-Holonomic Constraints ?

- Wheeled vehicles follow locally circular motion about the Instantaneous Center of Rotation (ICR)



Example of Ackerman steering principle



Locally circular motion

$$\varphi = \theta/2 \Rightarrow \textit{only 2 parameters } (\theta, \rho) \textit{ need to be estimated}$$

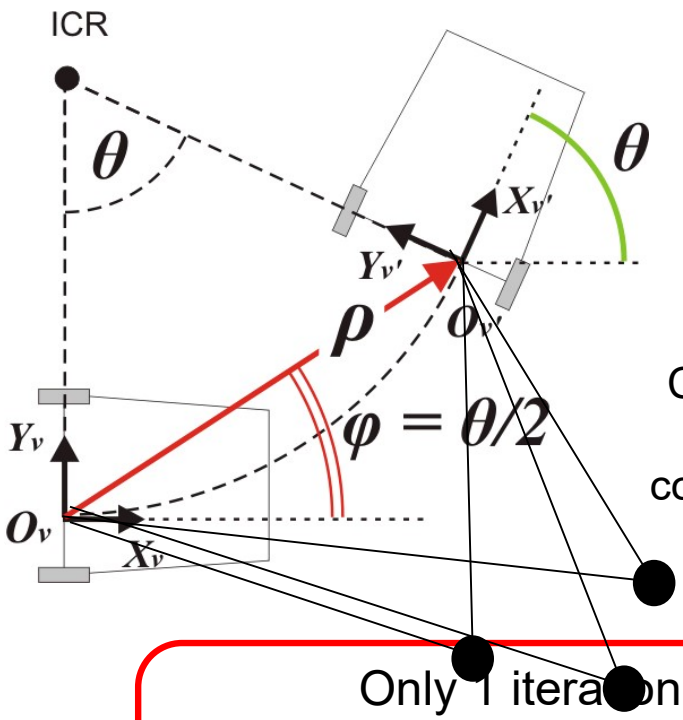
and therefore only 1 point is needed

This is the smallest parameterization possible and results in the most efficient algorithm for removing outliers

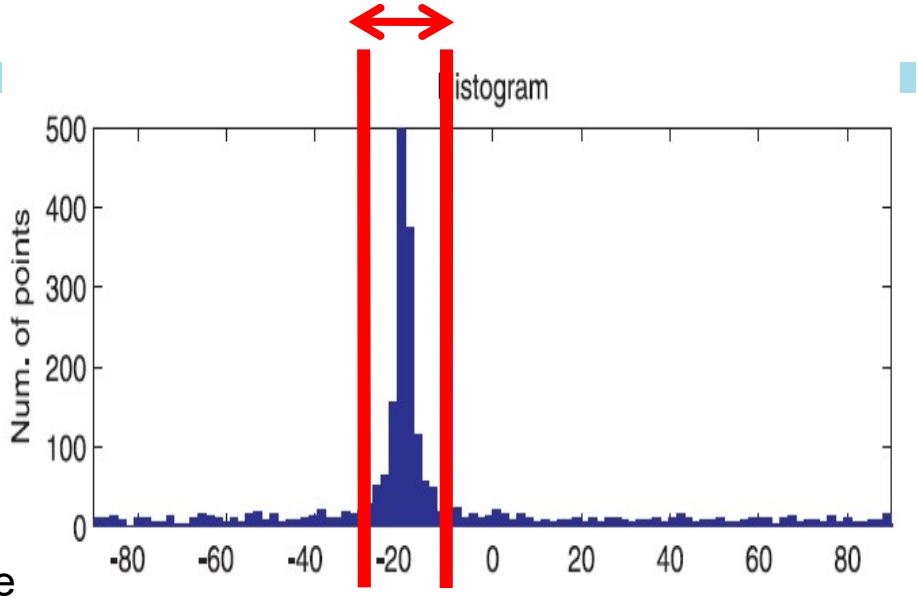
D. Scaramuzza. **1-Point-RANSAC Structure from Motion for Vehicle-Mounted Cameras by Exploiting Non-holonomic Constraints**. International Journal of Computer Vision, Volume 95, Issue 1, 2011

D. Scaramuzza. **Performance Evaluation of 1-Point-RANSAC Visual Odometry**. Journal of Field Robotics, Vol. 28, issue 5, 2011

1-Point RANSAC algorithm

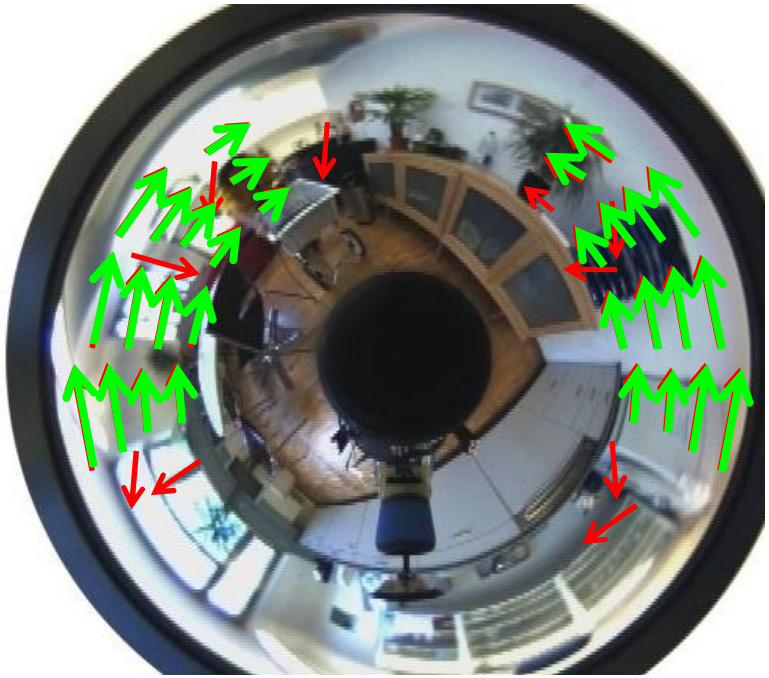


Compute θ for every point correspondence



Only 1 iteration

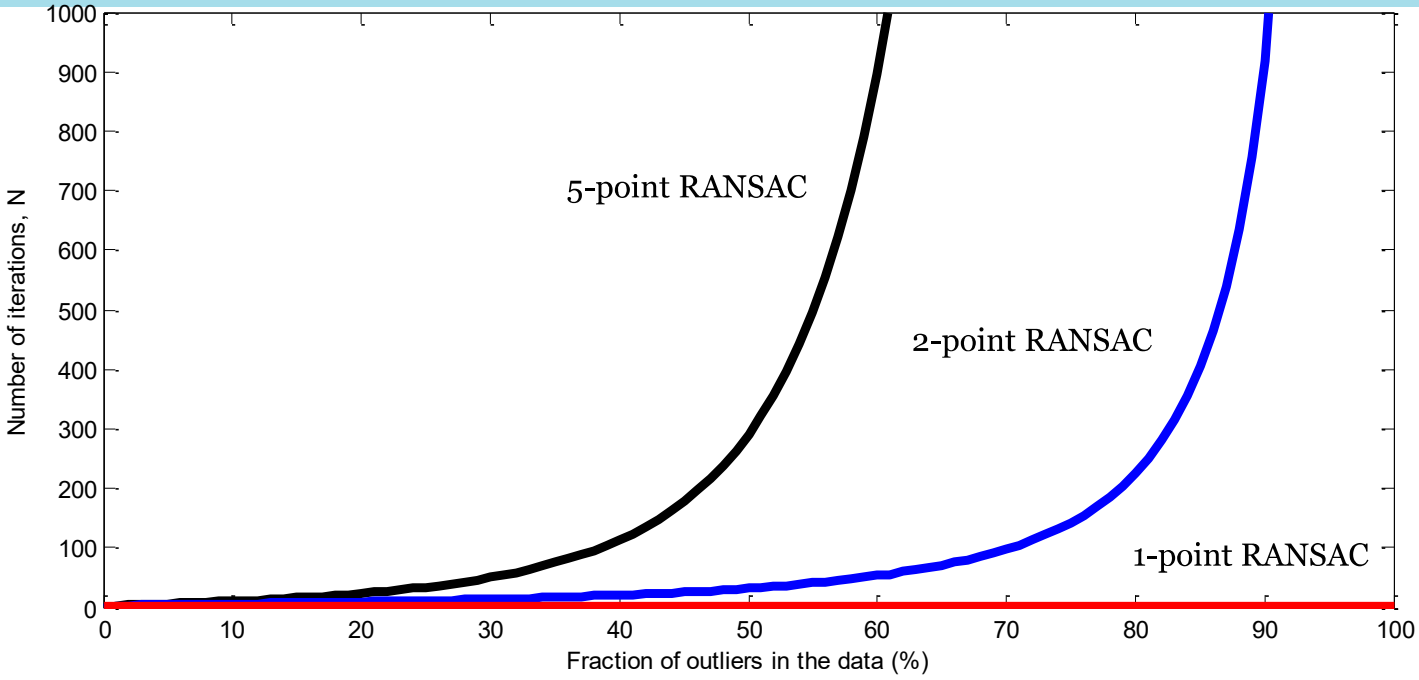
The most efficient algorithm for removing outliers, up to 800 Hz



1-Point RANSAC is ONLY used to find the inliers.
Motion is then estimated from them in 6DOF

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.or

Comparison of RANSAC algorithms



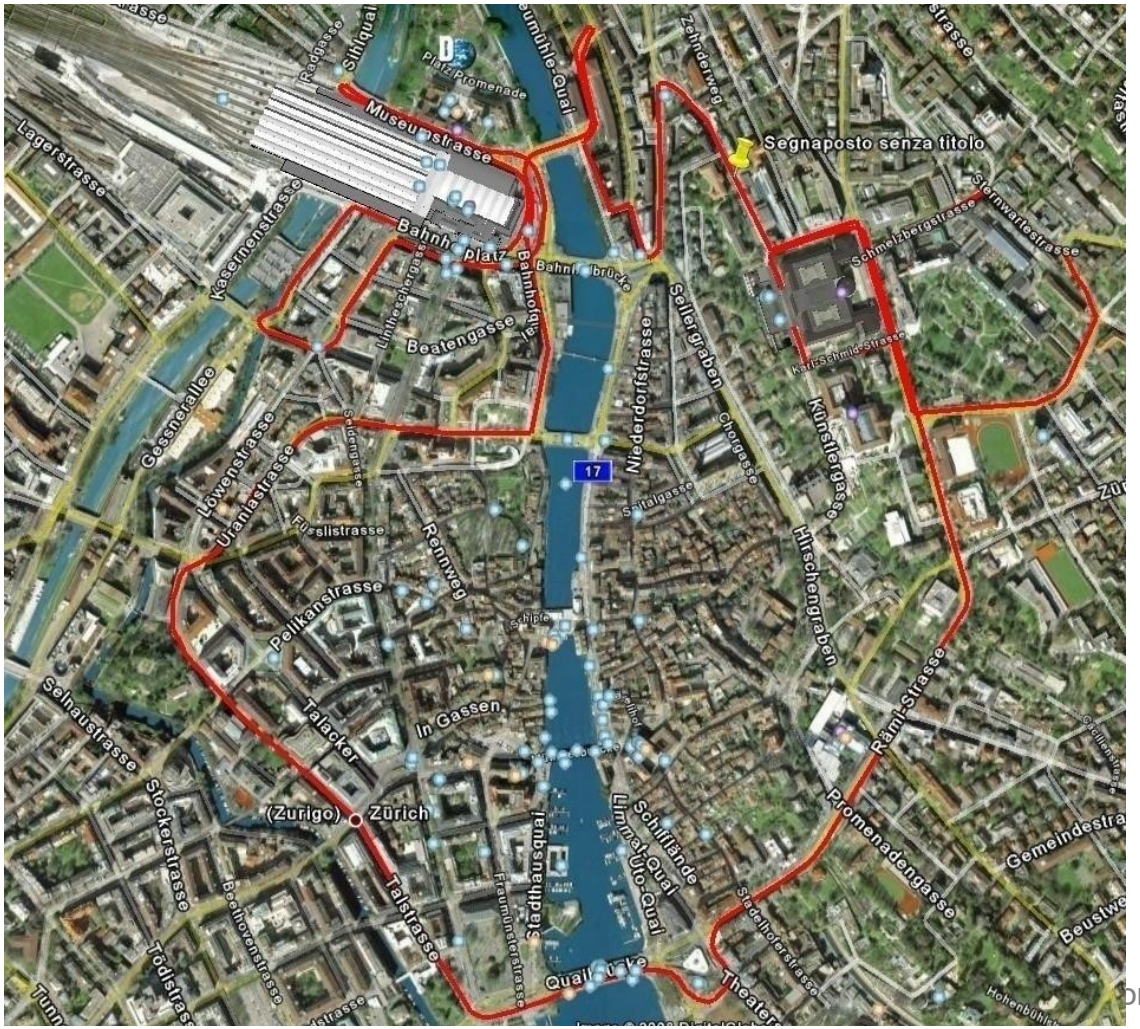
	5-Point RANSAC [Nister'03]	2-Point RANSAC [Ortin'01]	1-Point RANSAC [Scaramuzza, IJCV'11, JFR'11]
Number of iterations	**~1000**	**~100**	**1**

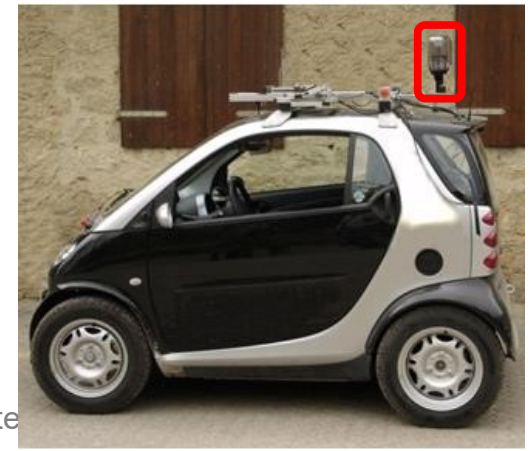
Our proposed method

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Visual Odometry: Results in a Urban Environment

- 15,000 images collected in Zurich during a over 25 Km path
- Image resolution: 640 x 480



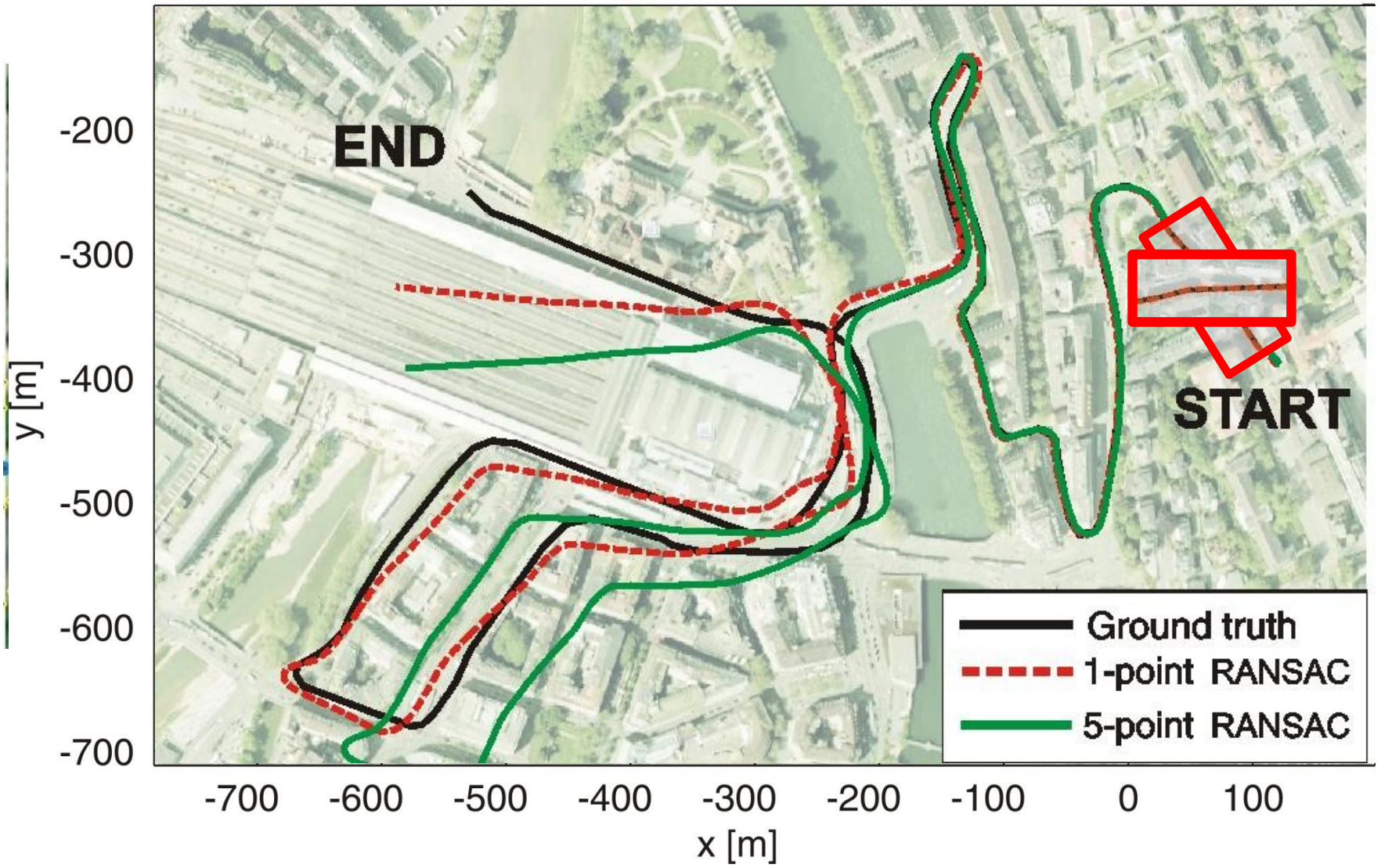


org - https://site

This video can be seen at
http://youtu.be/t7uKWZtUjCE

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Comparison between visual odometry and ground truth



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Considerations about RANSAC

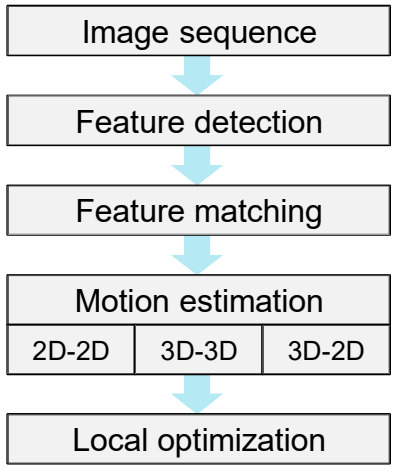
Is it really better to use minimal sets in RANSAC?

- If one is concerned with certain speed requirements, YES
- However, might not be a good choice if the image correspondences are very noisy: in this case, the motion estimated from a minimal set wil be inaccurate and will exhibit fewer inliers when tested on all other points
- Therefore, when the computational time is not a real concern and one deals with very noisy features, **using a non-minimal set may be better than using a minimal set**

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Outline

- Brief history of VO
- Problem formulation
- Camera modeling and calibration
- Motion estimation
- Robust estimation
- Error propagation
- Camera-pose optimization (bundle adjustment)
- Discussion



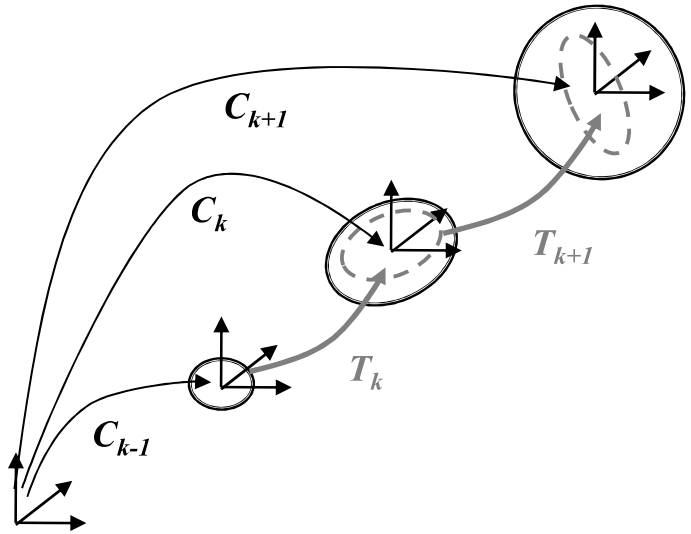
Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Error Propagation

- The uncertainty of the camera pose C_k is a combination of the uncertainty at C_{k-1} (black-solid ellipse) and the uncertainty of the transformation T_k (gray dashed ellipse)

- $C_k = f(C_{k-1}, T_k)$

- The combined covariance Σ_k is



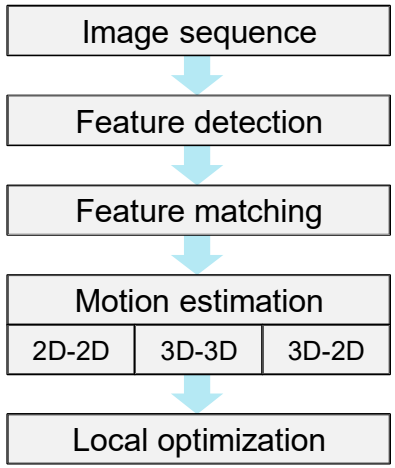
$$\begin{aligned}\Sigma_k &= J \begin{bmatrix} \Sigma_{k-1} & 0 \\ 0 & \Sigma_{k,k-1} \end{bmatrix} J^\top \\ &= J_{\vec{C}_{k-1}} \Sigma_{k-1} J_{\vec{C}_{k-1}}{}^\top + J_{\vec{T}_{k,k-1}} \Sigma_{k,k-1} J_{\vec{T}_{k,k-1}}{}^\top\end{aligned}$$

- The camera-pose uncertainty is always increasing when concatenating transformations. Thus, it is important to keep the uncertainties of the individual transformations small

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

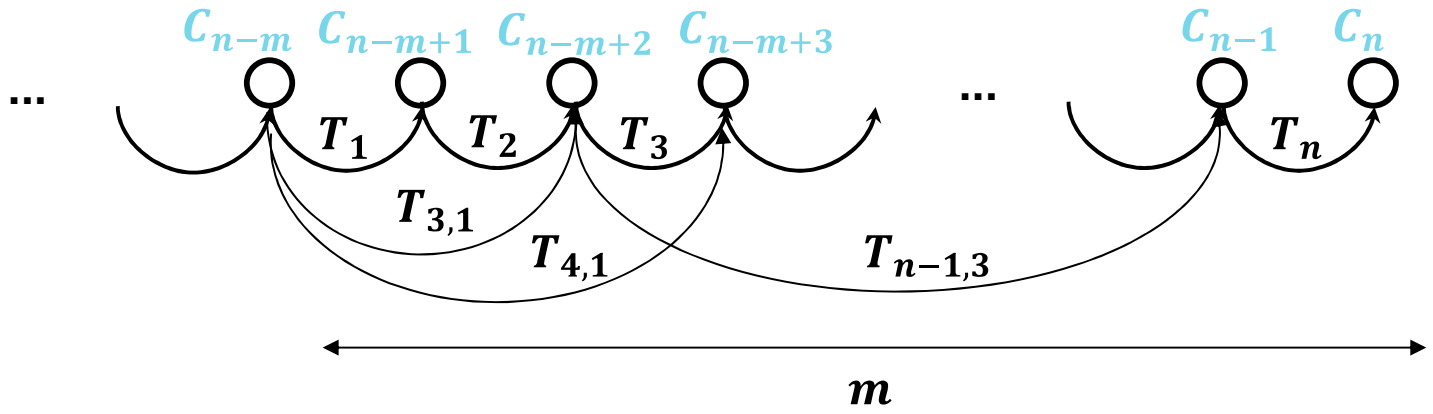
Outline

- Brief history of VO
- Problem formulation
- Camera modeling and calibration
- Motion estimation
- Robust estimation
- Error propagation
- Camera-pose optimization (bundle adjustment)
- Discussion



Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Windowed Camera-Pose Optimization



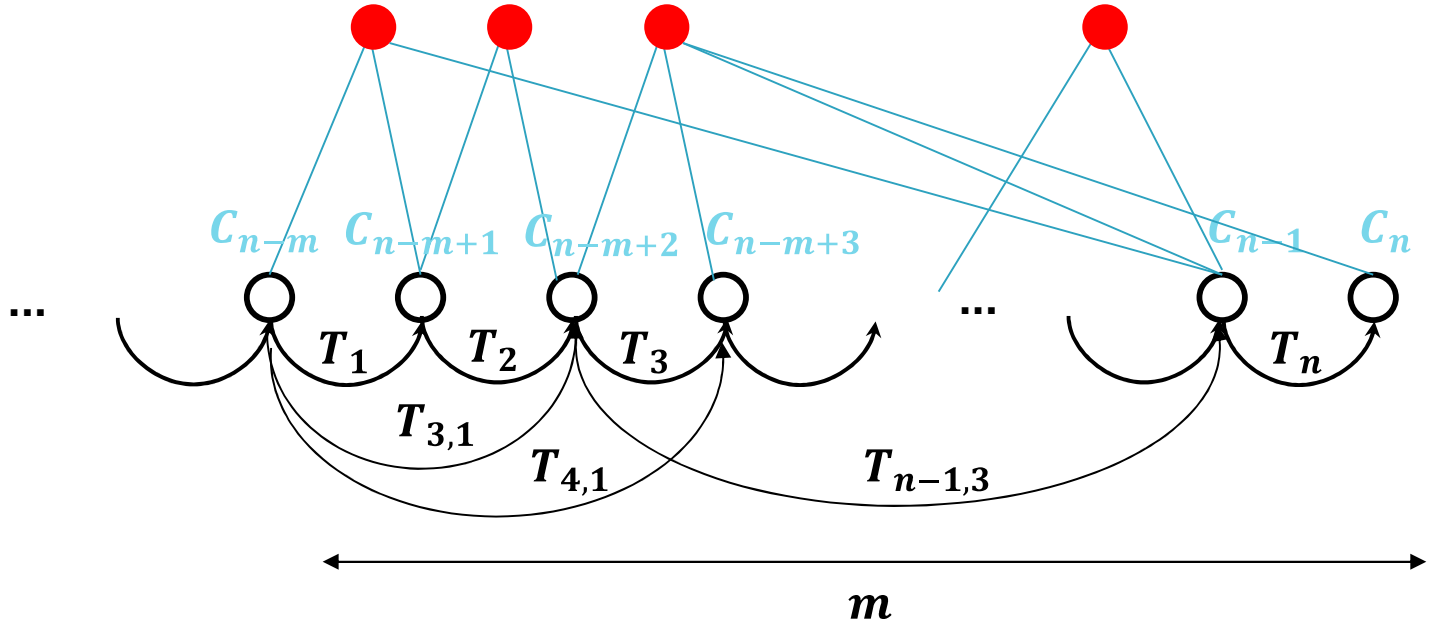
- So far we assumed that the transformations are between consecutive frames
- Transformations can be computed also between non-adjacent frames $T_{e_{ij}}$ and can be used as additional constraints to improve cameras poses by minimizing the following

$$\sum_{e_{ij}} \|C_i - T_{e_{ij}} C_j\|^2$$

- For efficiency, only the last m keyframes are used
- Levenberg-Marquadt can be used

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Windowed Bundle Adjustment (BA)



- Similar to pose-optimization but it also optimizes 3D points

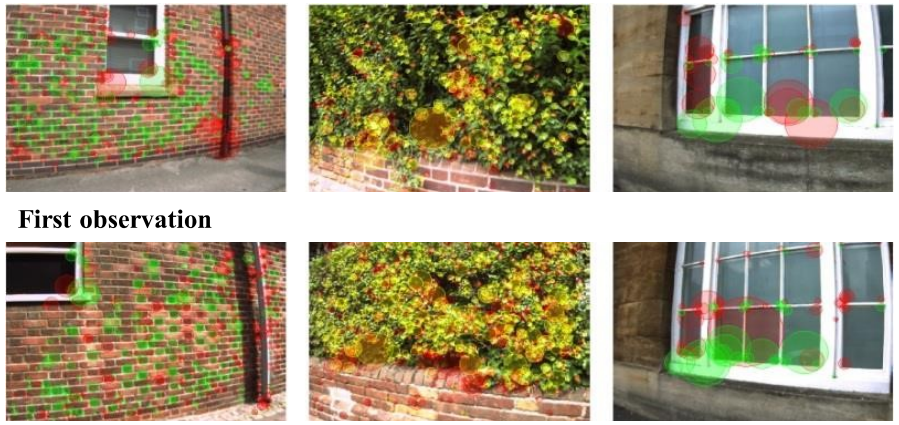
$$\arg\min_{X^i, C_k} \sum_{i,k} \| p_k^i - g(X^i, C_k) \|^2$$

- In order to not get stuck in local minima, the initialization should be close the minimum
- Levenberg-Marquadt can be used

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Loop Detection

- Loop constraints are very valuable constraints for pose graph optimization
- These constraints form graph edges between nodes that are usually far apart and between which large drift might have been accumulated.
- Events like reobserving a landmark after not seeing it for a long time or coming back to a previously-mapped area are called loop detections
- Loop constraints can be found by evaluating visual similarity between the current camera images and past camera images.
- Visual similarity can be computed using global image descriptors or local image descriptors (see lecture about Visual SLAM)



First observation

Second observation after a loop

Image courtesy of Cummins & Newman, IJRR'08

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Considerations about Bundle Adjustment

- Windowed BA reduces the drift compared to 2-view VO because incorporates constraints between several frames
- More precise than camera-pose optimization
- The choise of the window size *m* is governed by computational reasons
- The computational complexity of BA is $O\left((qN + lm)^3\right)$ with N being the number of points, m the number of poses, and q and m the number of parameters for points and camera poses

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Improving the Accuracy of VO

- Other sensors can be used such as
  - IMU (called inertial VO)
  - Compass
  - GPS
  - Laser
- An IMU combined with a single camera allows the estimation of the absolute scale. Why?
- Make sure that you have many points (thoudsands) which cover the image uniformly

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

VO Applications

VO has successfully been applied within various technological fields

- Space exploration:
  - Planetary lander furing descent phase
  - Spirit and Opportunity Mars-exploration rovers
    - Since 2004, used VO in addition to dead-reckoning for about 6 Km
    - Especially in preence of wheel slip



- MAV navigation
  - European project SFLY
  - Vision-based MAVs at the Robotics and Perception Group (see http://rpg.ifi.uzh.ch/research_mav.html)

- Underwater vehicles

- Automotive industry

The sFly video can be seen at http://youtu.be/_-p08o_oTO4

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

VO Applications

World-first mouse scanner

Currently distributed by LG: *SmartScan LG LSM100*



This video can be seen at
http://youtu.be/A4NGXFv27AE





Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Software and Dataset

SOFTWARE AND DATASETS

Author	Description	Link
Willow Garage	OpenCV: A computer vision library maintained by Willow Garage. The library includes many of the feature detectors mentioned in this tutorial (e.g., Harris, KLT, SIFT, SURF, FAST, BRIEF, ORB). In addition, the library contains the basic motion-estimation algorithms as well as stereo-matching algorithms.	http://opencv.willowgarage.com
Willow Garage	ROS (Robot Operating System): A huge library and middleware maintained by Willow Garage for developing robot applications. Contains a visual-odometry package and many other computer-vision-related packages.	http://www.ros.org
Willow Garage	PCL (Point Cloud Library): A 3D-data-processing library maintained from Willow Garage, which includes useful algorithms to compute transformations between 3D-point clouds.	http://pointclouds.org
Henrik Stewenius et al.	5-point algorithm: An implementation of the 5-point algorithm for computing the essential matrix.	http://www.vis.uky.edu/~stewe/FIVEPOINT/
Changchang Wu et al.	SiftGPU: Real-time implementation of SIFT.	http://cs.unc.edu/~ccwu/siftgpu
Nico Cornelis et al.	GPUSurf: Real-time implementation of SURF.	http://homes.esat.kuleuven.be/~ncorneli/gpusurf
Christopfer Zach	GPU-KLT: Real-time implementation of the KLT tracker.	http://www.inf.ethz.ch/personal/chzach/opensource.html
Edward Rosten	Original implementation of the FAST detector.	http://www.edwardrosten.com/work/fast.html

Copyright of Davide Scaramuzza - davide.scaramuzza@ieee.org - https://sites.google.com/site/scarabotix/

Software and Dataset

Michael Calonder	Original implementation of the BRIEF descriptor.	http://cvlab.epfl.ch/software/brief/
Leutenegger et al.	BRISK feature detector.	http://www.asl.ethz.ch/people/lestefan/personal/BRISK
Jean-Yves Bouguet	Camera Calibration Toolbox for Matlab.	http://www.vision.caltech.edu/bouguetj/calib_doc
Davide Scaramuzza	OCamCalib: Omnidirectional Camera Calibration Toolbox for MATLAB.	https://sites.google.com/site/scarabotix/ocamcalib-toolbox
Christopher Mei	Omnidirectional Camera Calibration Toolbox for MATLAB	http://homepages.laas.fr/~cmei/index.php/Toolbox
Mark Cummins	FAB-MAP: Visual-word-based loop detection.	http://www.robots.ox.ac.uk/~mjc/Software.htm
Friedrich Fraundorfer	Vocsearch: Visual-word-based place recognition and image search.	http://www.inf.ethz.ch/personal/fraundof/page2.html
Manolis Lourakis	SBA: Sparse Bundle Adjustment	http://www.ics.forth.gr/~lourakis/sba
Christopher Zach	SSBA: Simple Sparse Bundle Adjustment	http://www.inf.ethz.ch/personal/chzach/opensource.html
Rainer Kuemmerle et al.	G2O: Library for graph-based nonlinear function optimization. Contains several variants of SLAM and bundle adjustment.	http://openslam.org/g2o
RAWSEEDS EU Project	RAWSEEDS: Collection of datasets with different sensors (lidars, cameras, IMUs, etc.) with ground truth.	http://www.rawseeds.org
SFLY EU Project	SFLY-MAV dataset: Camera-IMU dataset captured from an aerial vehicle with Vicon data for ground truth.	http://www.sfly.org
Davide Scaramuzza	ETH OMNI-VO: An omnidirectional-image dataset captured from the roof of a car for several kilometers in a urban environment. MATLAB code for visual odometry is provided.	http://sites.google.com/site/scarabotix